

ASSIGNMENT 1
CMOS ANALOG IC DESIGN
BILAVATH PREM SINGH (25210098)

TASK 1

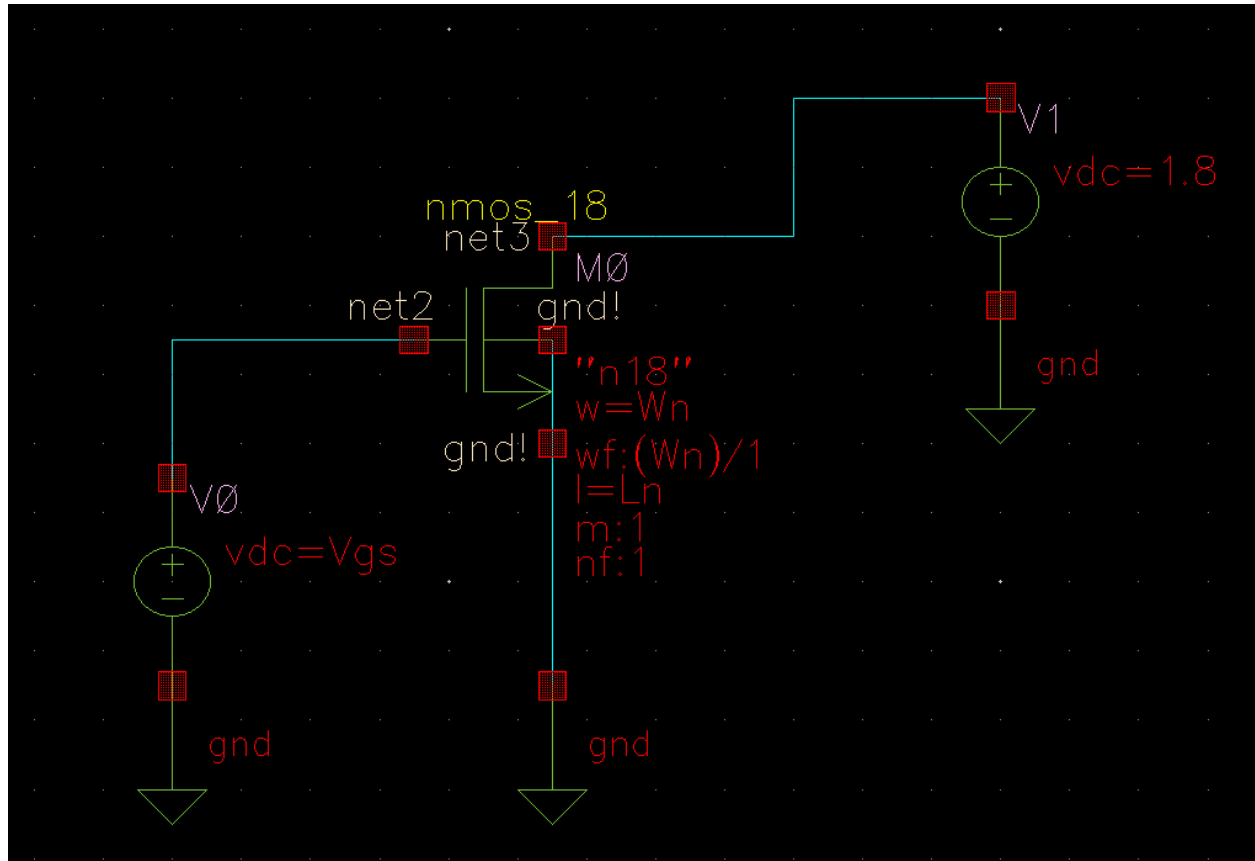


FIG:-1.0 SCHEMATIC CIRCUIT OF NMOS

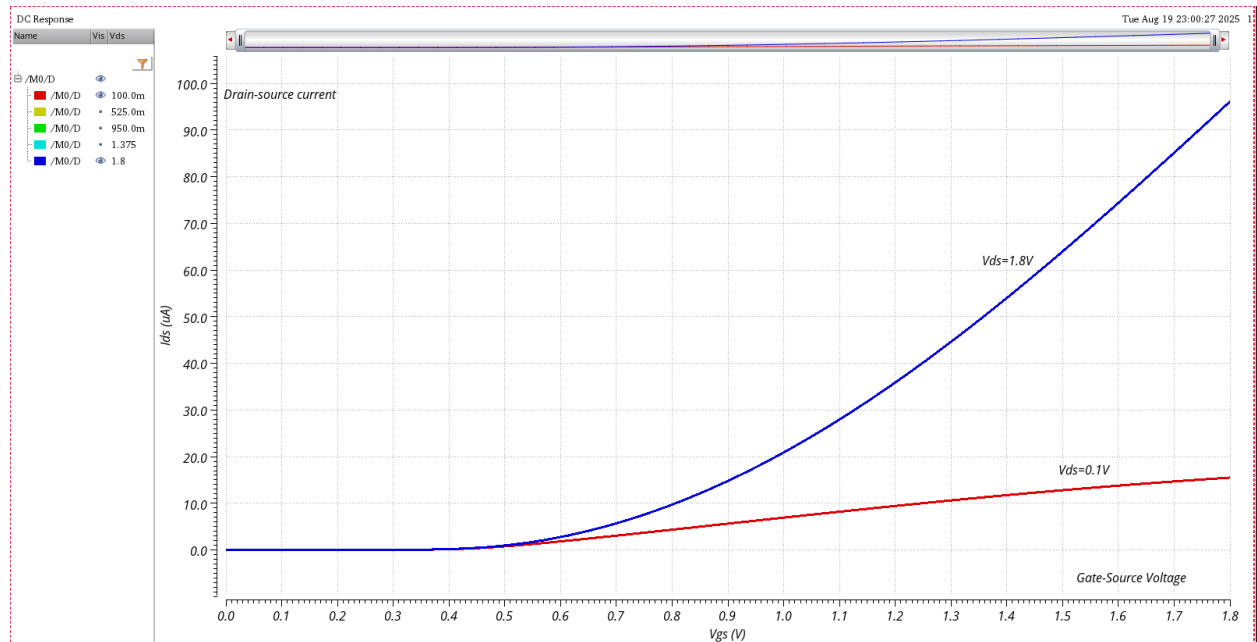


FIG:-1.1 NMOS TRANSFER CHARACTERISTICS

Transfer Characteristics:- V_{gs} VS I_{ds}

NMOS: At low V_{gs} , the NMOS is off (no current flows). As V_{gs} increases beyond the threshold voltage (V_{th} , typically around 0.4-0.5V for NMOS), the transistor turns on, and the drain current (I_{ds}) starts to increase. At $V_{ds} = 100$ mV (a small voltage), the transistor operates in the linear region, and I_{ds} increases almost linearly with V_{gs} .

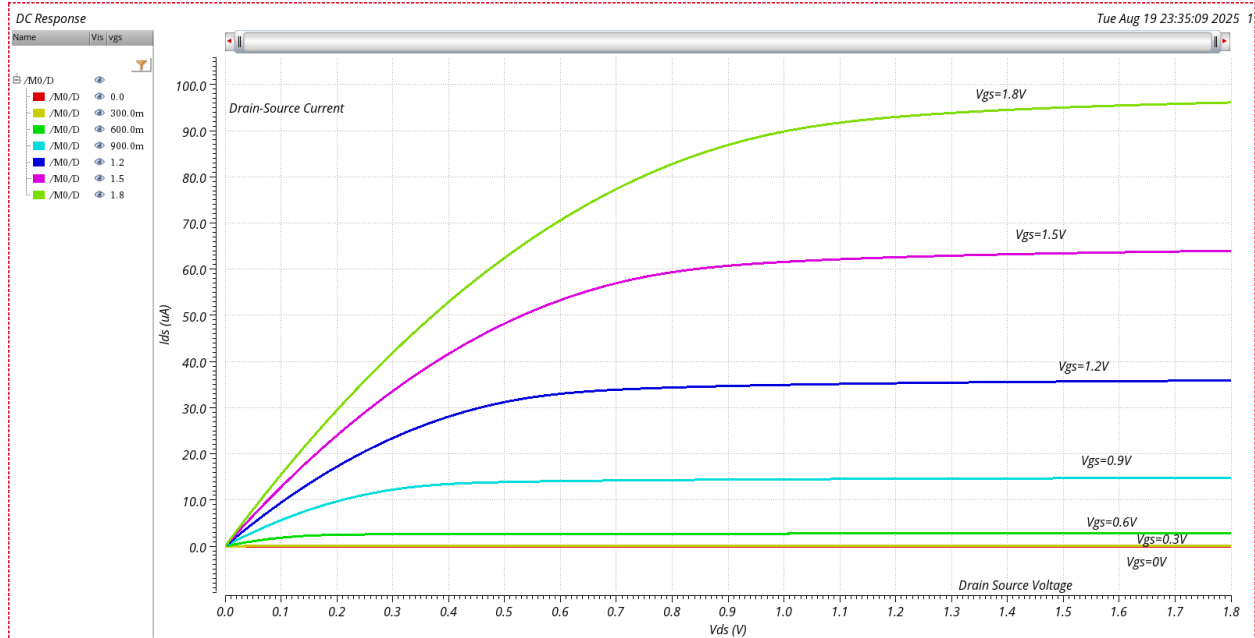
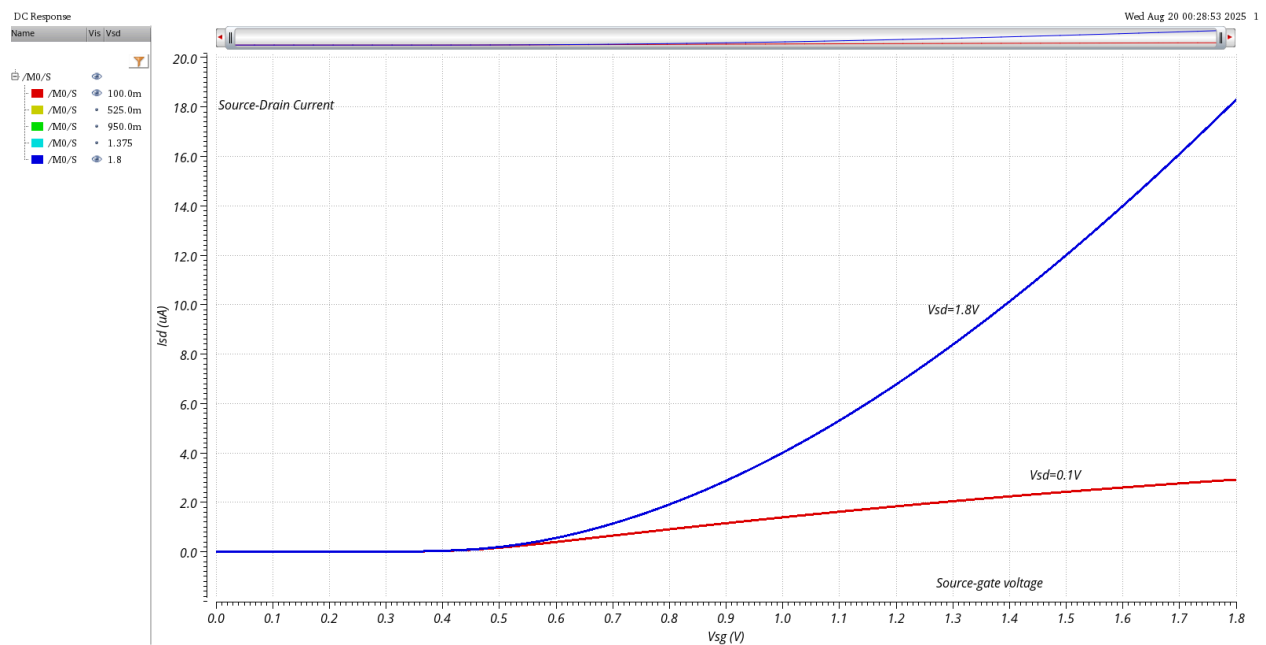
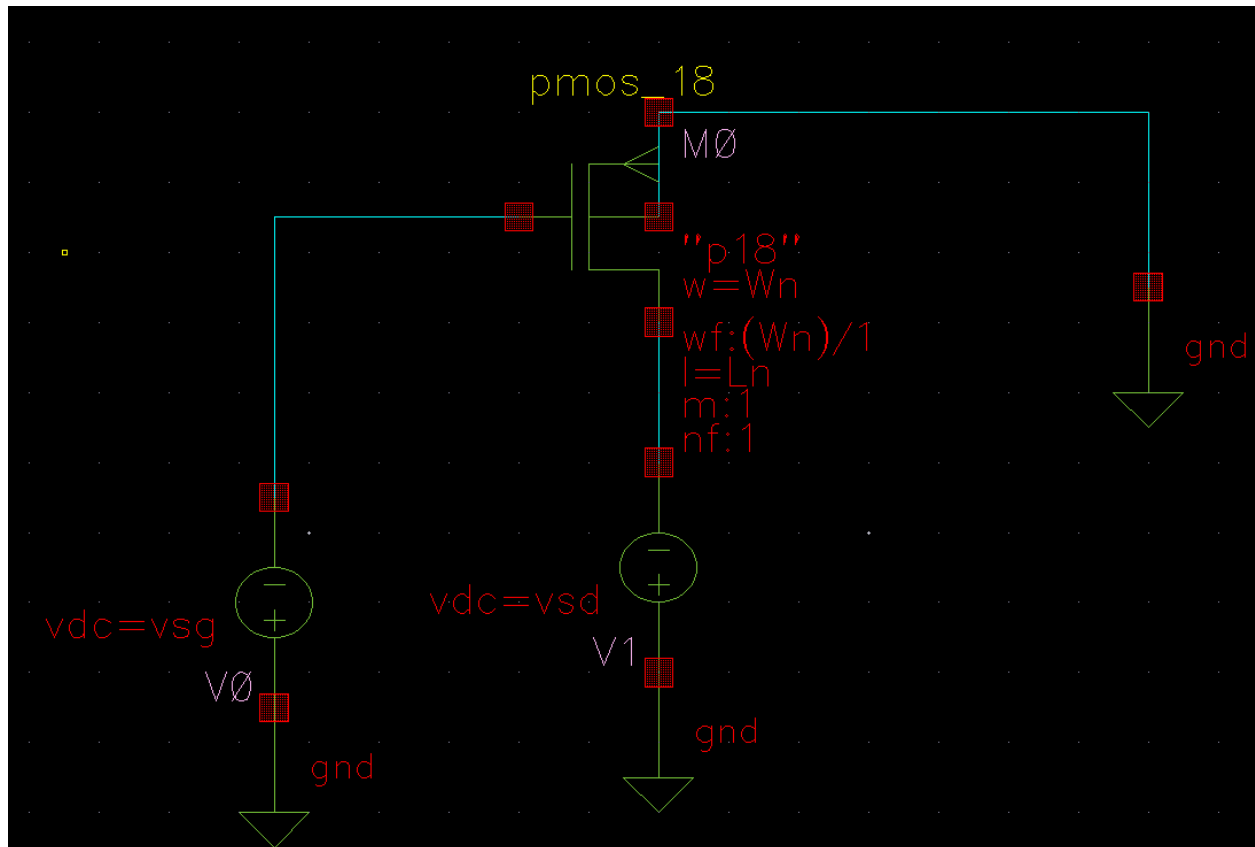


FIG:-1.2 NMOS OUTPUT CHARACTERISTICS

NMOS: As V_{GS} increases to 1.8V, the NMOS transitions from the linear region to the saturation region when $V_{GS} - V_{th} < V_{DS}$. The current (I_{DS}) will saturate at higher V_{DS} values for a given V_{GS} . You'll observe a curve where I_{DS} Vs V_{DS} increases. With V_{GS} increasing in steps of 0.3 the I_{DS} current also increases



PMOS: The PMOS works oppositely. It turns on when V_{gs} becomes sufficiently negative (below its threshold voltage, typically -0.4 to -0.5V). At $V_{ds} = 100 \text{ mV}$ (with V_s more positive than V_d), the current will flow when V_{gs} is negative enough, and I_{sd} will increase as V_{gs} becomes more negative.

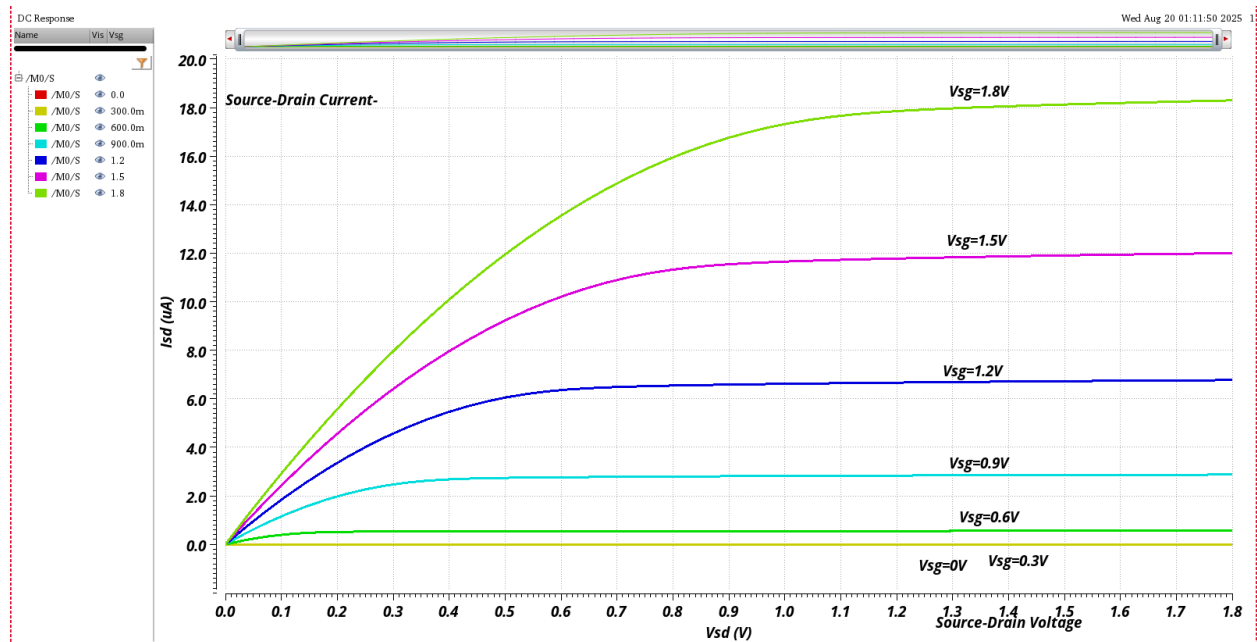


FIG:-1.5 PMOS OUTPUT CHARACTERISTICS

PMOS: For PMOS, $V_{sd} = 1.8\text{V}$ means the source is at a higher voltage than the drain. The PMOS will saturate when $|V_{gs}| - |V_{tp}| < V_{sd}$. The current will increase with more negative V_{sg} and then level off in saturation. Here in this graph I have taken current from the drain terminal.

TASK 2

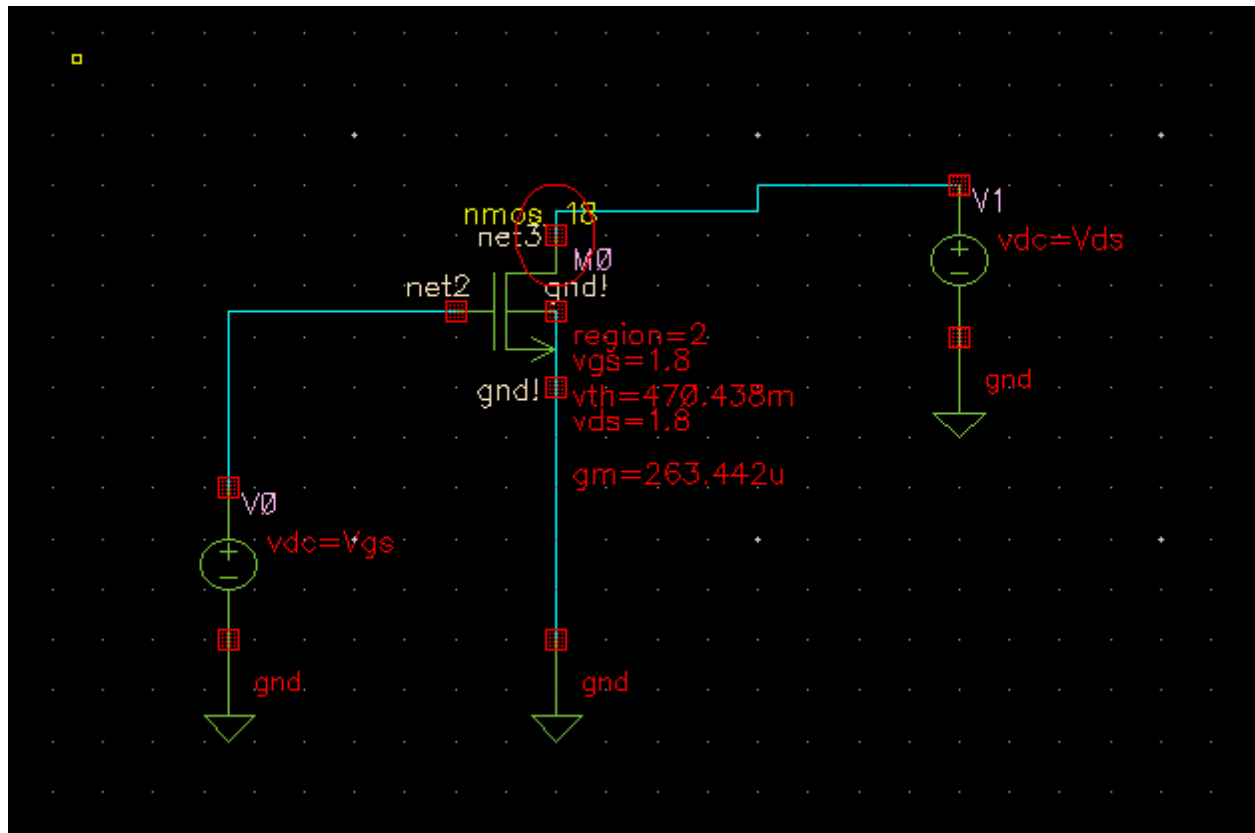


FIG:-2.0 NMOS SCHEMATIC CIRCUIT FOR SATURATION

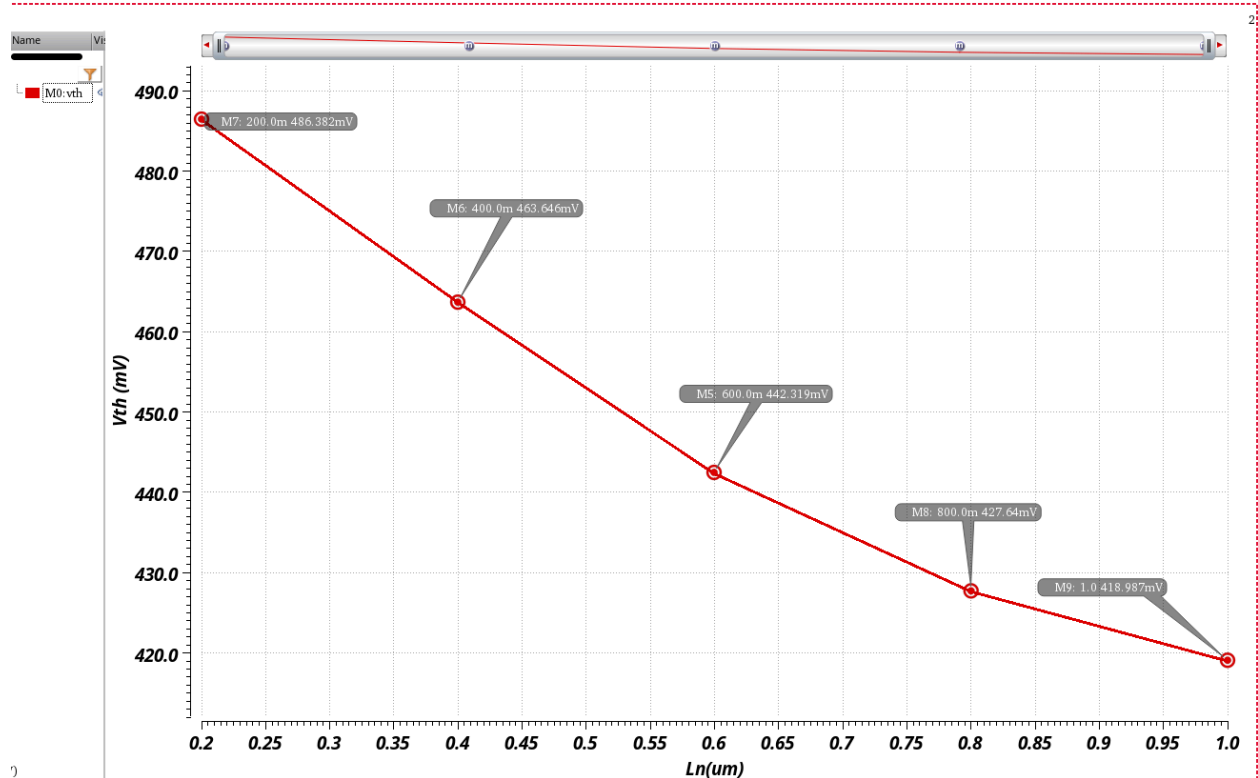


FIG:-2.1 THRESHOLD VOLTAGE AS FUNCTION OF GATE LENGTH

$$V_{T0} = \phi_{MS} + 2\phi_f + \gamma\sqrt{2\phi_f}$$

The threshold voltage (V_{th}) increases slightly as the gate length (L_g) increases from 200 nm to 1000 nm. This happens because a longer channel reduces short-channel effects, making it harder to turn on the transistor, thus raising V_{th} . For both NMOS and PMOS, I saw a gradual rise in V_{th} with longer L_g .

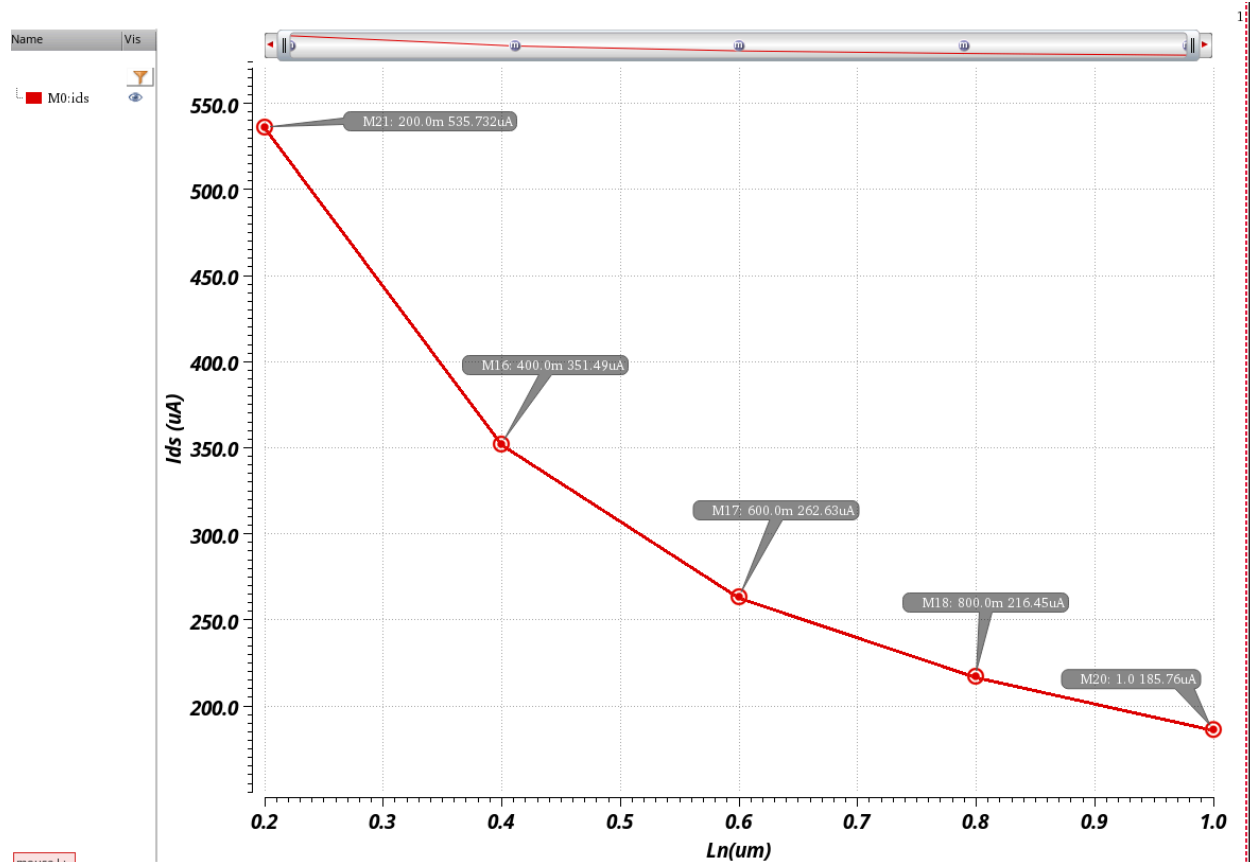


FIG:-2.2 DRAIN-SOURCE CURRENT AS FUNCTION OF GATE LENGTH

$$I_{DS} \approx \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{TH})^2 (1 + \lambda V_{DS})$$

The drain current (I_{DS}) in the saturation region decreases as L_g increases. A longer channel increases resistance, reducing the current flow. I have noticed that I_{DS} is highest for the 200 nm length and lowest for the 1000 nm length for both NMOS and PMOS, with a steady drop as L_g grows.

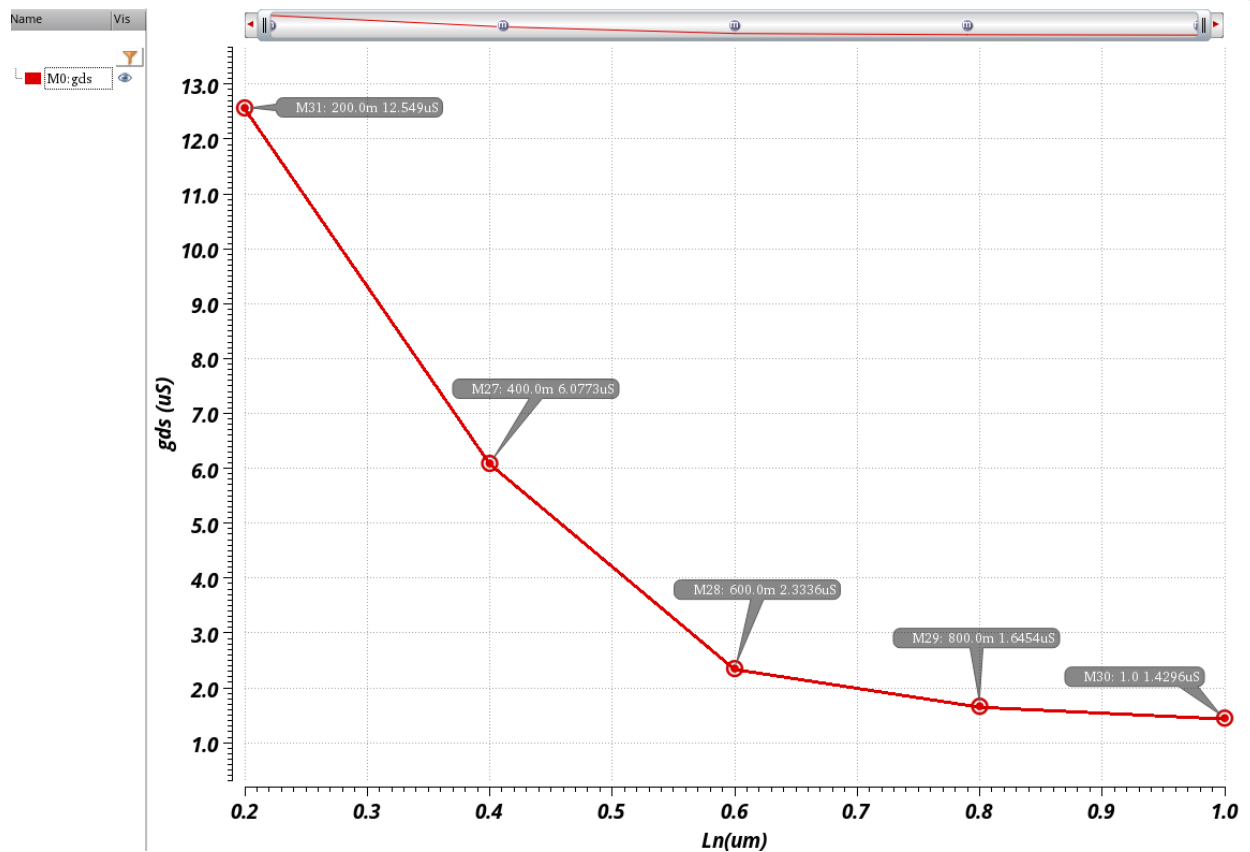


FIG:-2.3 DRAIN-SOURCE CONDUCTANCE AS FUNCTION OF GATE LENGTH

$$\begin{aligned}
 g_{ds} &= \frac{\partial I_{ds}}{\partial V_{ds}} = \mu_n C_{ox} \frac{W}{L} [V_{gs} - V_{th} - V_{ds}] \\
 &= k \cdot \frac{W}{L} [V_{gs} - V_{th} - V_{ds}] \\
 &= \beta [V_{gs} - V_{th} - V_{ds}]
 \end{aligned}$$

for 1000 nm compared to 200 nm. So saw a declining trend for both NMOS and PMOS. The output conductance (g_{ds}), which shows how much I_{ds} changes with V_{ds} , decreases as L_g increases. Longer channels reduce the impact of drain voltage on current, so g_{ds} is lower.

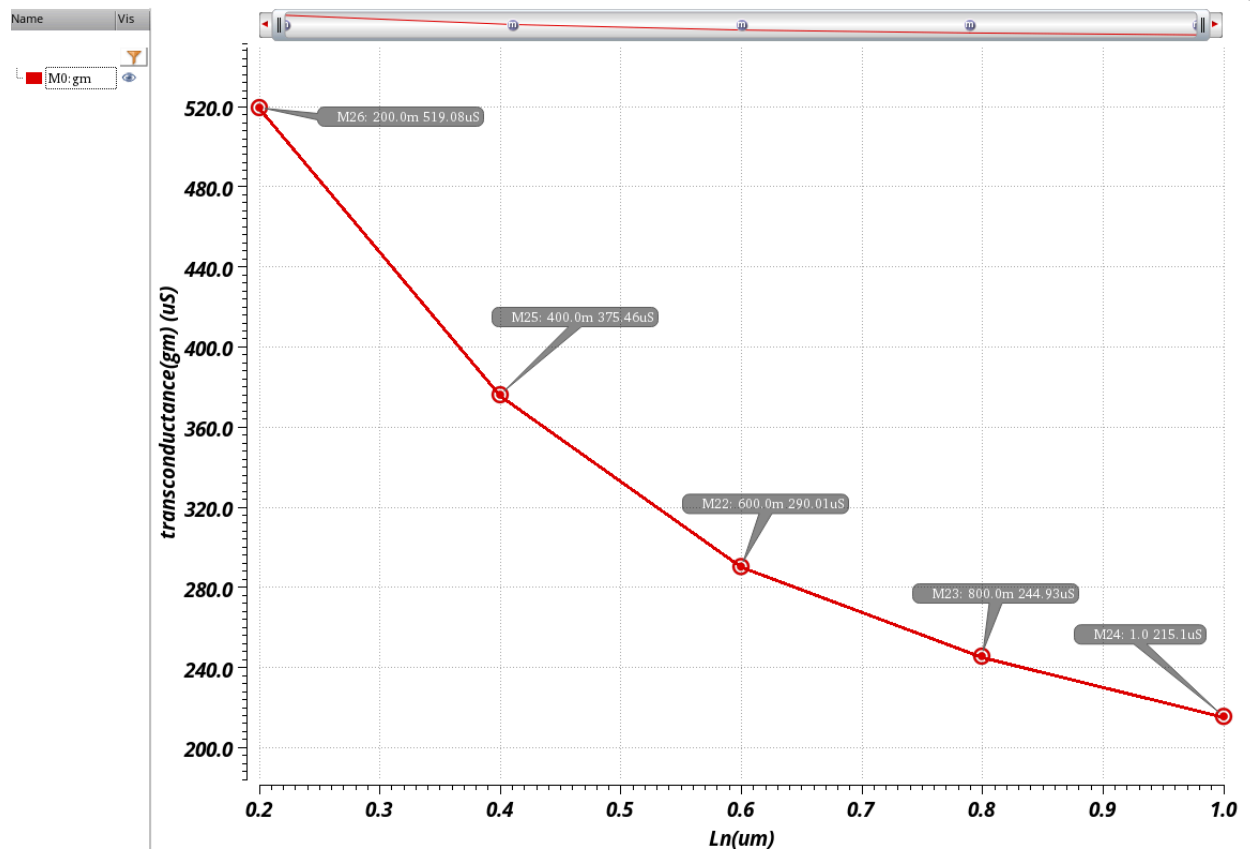


FIG:-2.4 TRANSCONDUCTANCE (gm) AS FUNCTION OF GATE LENGTH(NMOS)

$$\begin{aligned}
 g_m &= \left. \frac{\partial i_D}{\partial v_{GS}} \right|_{v_{GS}=V_{GS}} = \frac{i_D}{v_{GS}} \\
 &= (\mu_n C_{ox}) \left(\frac{W}{L} \right) (V_{GS} - V_{th}) = (\mu_n C_{ox}) \left(\frac{W}{L} \right) V_{OV} \\
 &= \sqrt{2(\mu_n C_{ox}) \left(\frac{W}{L} \right) I_D} \\
 &= \frac{2I_D}{V_{OV}}
 \end{aligned}$$

The transconductance (gm), which measures how effectively the transistor controls current with Vgs, decreases as Lg increases. A longer channel reduces the ability to

modulate current, so gm is higher for shorter lengths (like 200 nm) and lower for longer lengths (like 1000 nm) in both NMOS and PMOS.

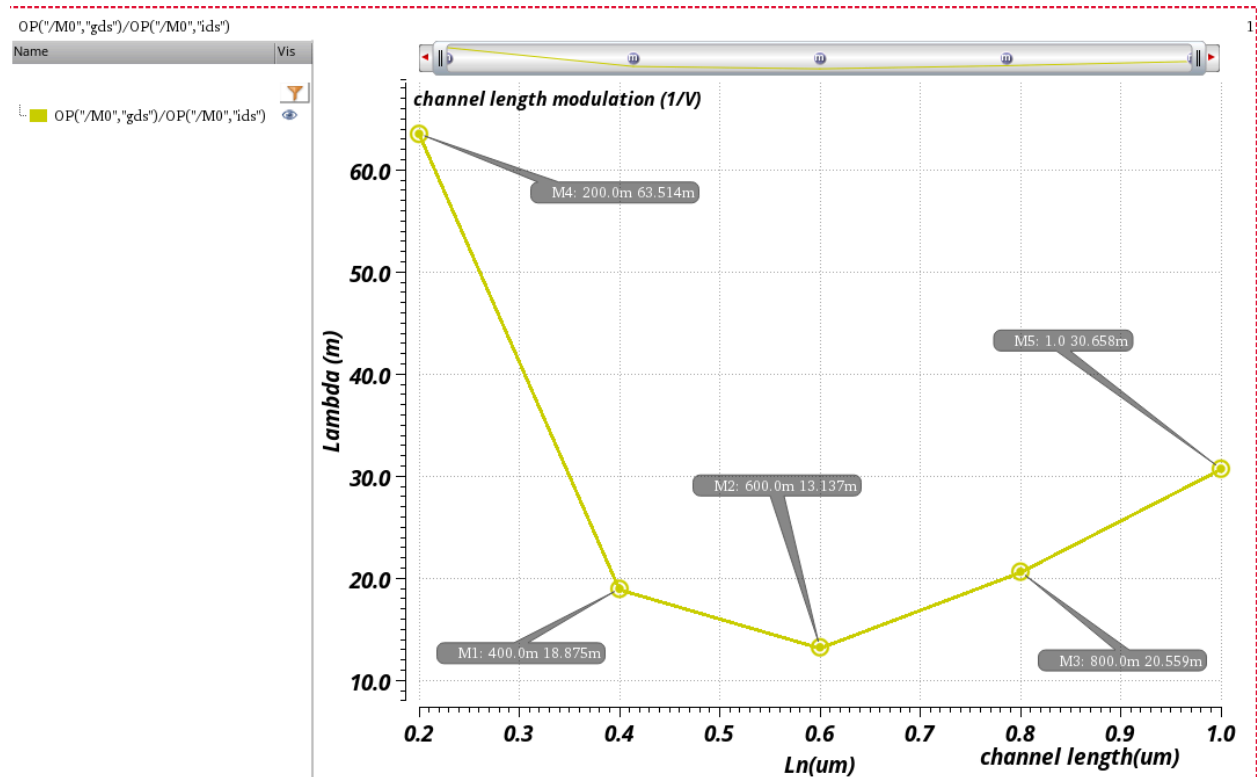


FIG:-2.5 LAMBDA AS FUNCTION OF GATE LENGTH (NMOS)

The channel-length modulation parameter (λ), which affects how much I_{ds} increases with V_{ds} in saturation, decreases as L_g increases. Shorter channels (200 nm) have a higher λ due to stronger modulation effects, while longer channels (600 nm) show a smaller λ for both NMOS and PMOS

Ln(um)	Vtn(mV)	Ids(uA)	gm(uS)	gds(uS)	lambda(1/V)
0.2	486.382	535.732	519.08	12.549	63.514
0.4	463.646	351.49	375.46	6.07	18.875
0.6	442.319	262.63	290.01	2.3	13.137
0.8	427.64	216.45	244.93	1.64	20.559
1.0	418.987	185.76	215.1	1.42	30.658

NMOS IN SATURATION

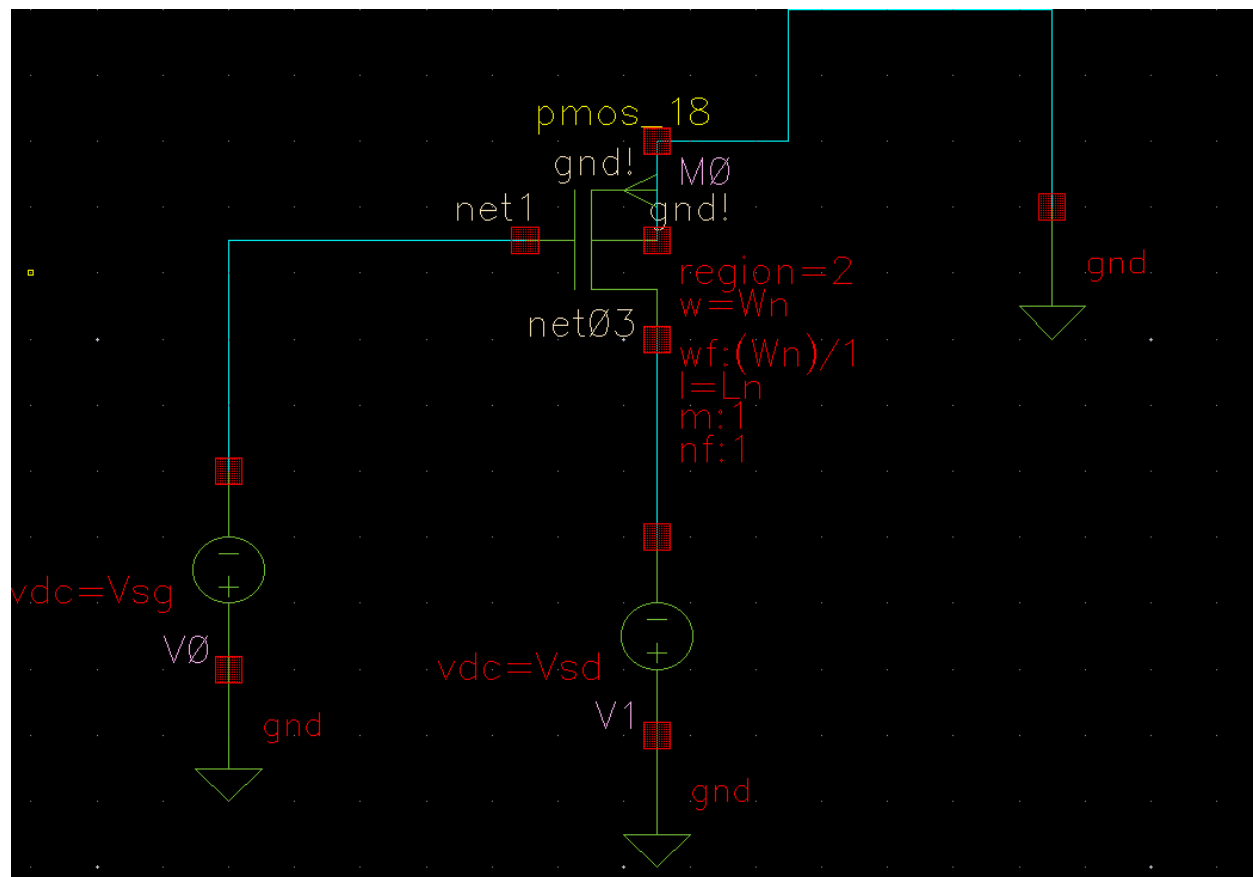


FIG:-2.6 PMOS SCHEMATIC CIRCUIT FOR SATURATION

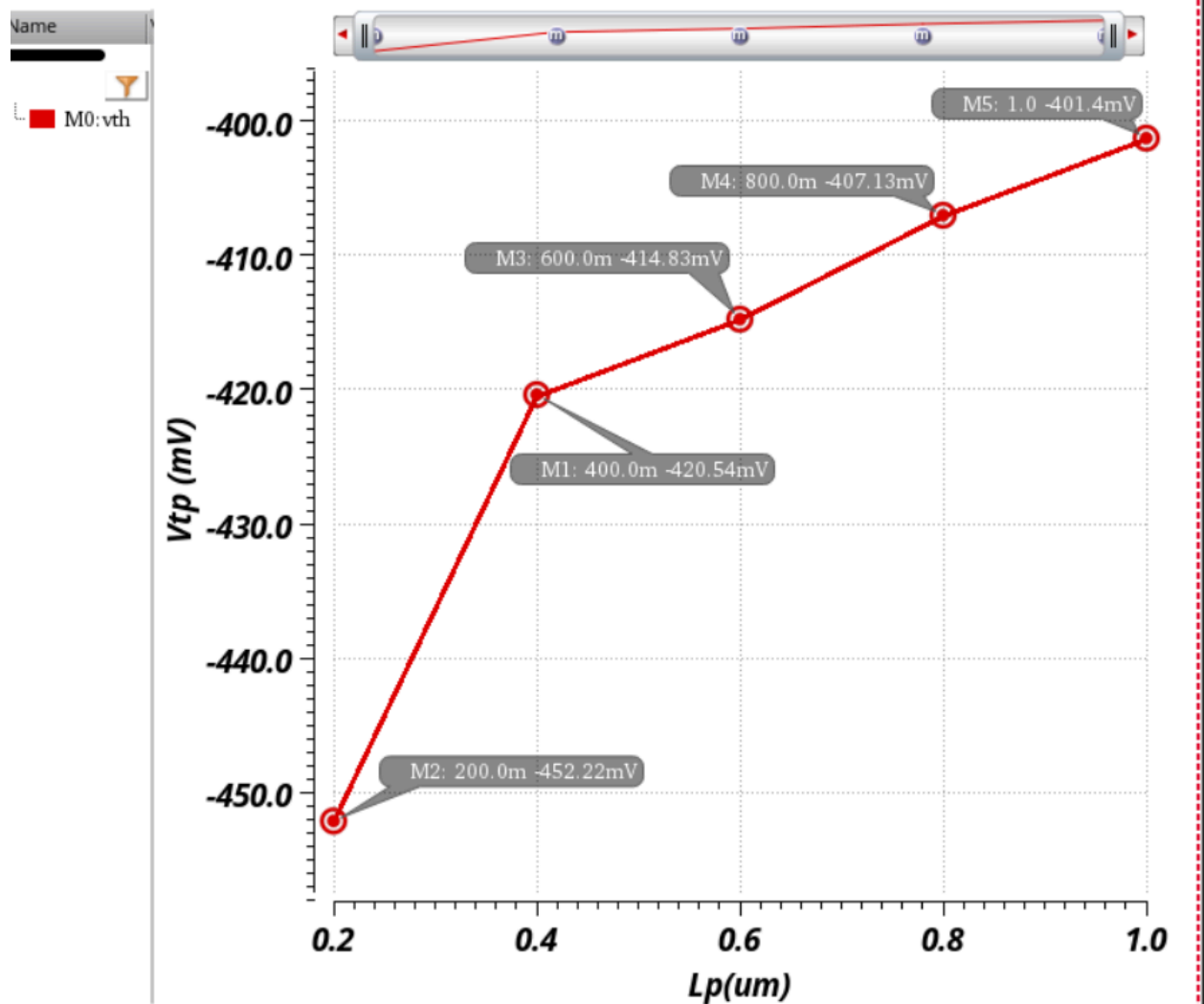


FIG:-2.7 THRESHOLD VOLTAGE AS FUNCTION OF GATE LENGTH (PMOS)

Lp(um)	Vtp(mV)
0.2	-452.22
0.4	-420.54
0.6	-414.8
0.8	-407.13
1	-401.4

Here we can see the dependence of channel length on threshold voltage i.e. increasing with length

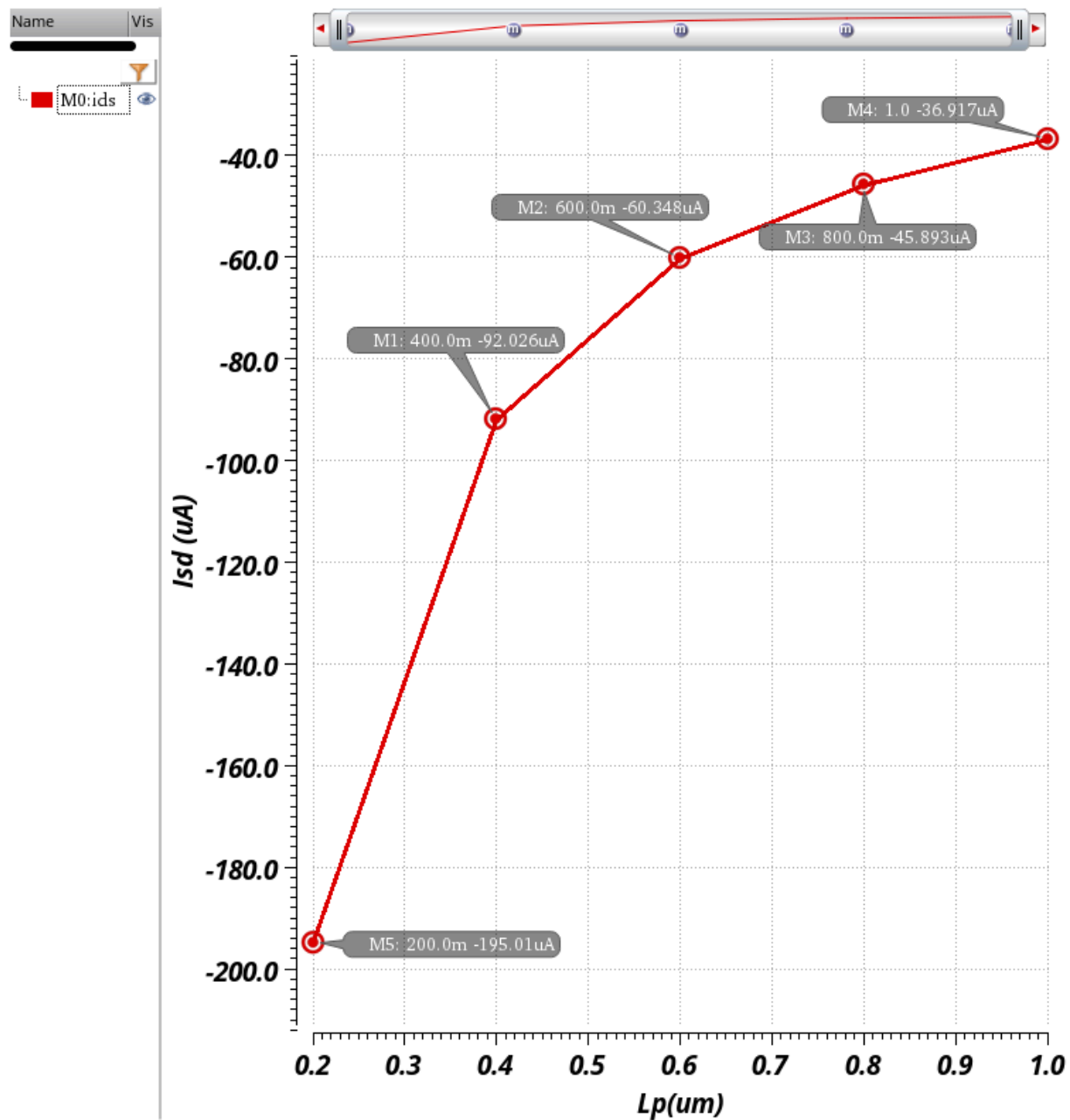


FIG:- 2.8 SOURCE-DRAIN CURRENT AS FUNCTION OF GATE LENGTH (PMOS)

$L_p(\mu m)$	$I_{sd}(\mu A)$
0.2	-195.01
0.4	-92.02
0.6	-60.348
0.8	-45.81

As length increases current decreases because its inversely proportional to length and we can observe the trend

2

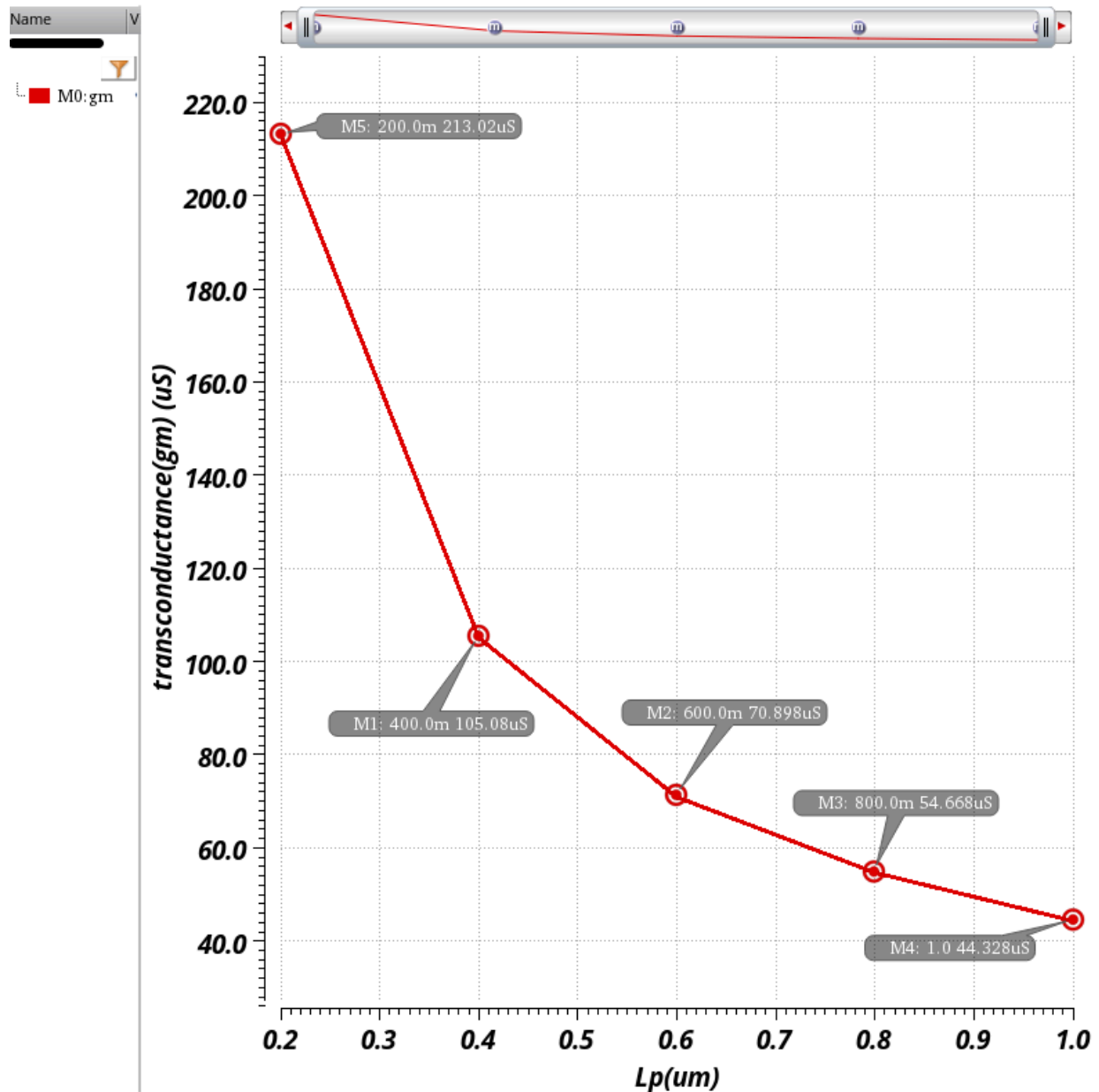


FIG:- 2.9 TRANSCONDUCTANCE AS FUNCTION OF GATE LENGTH (PMOS)

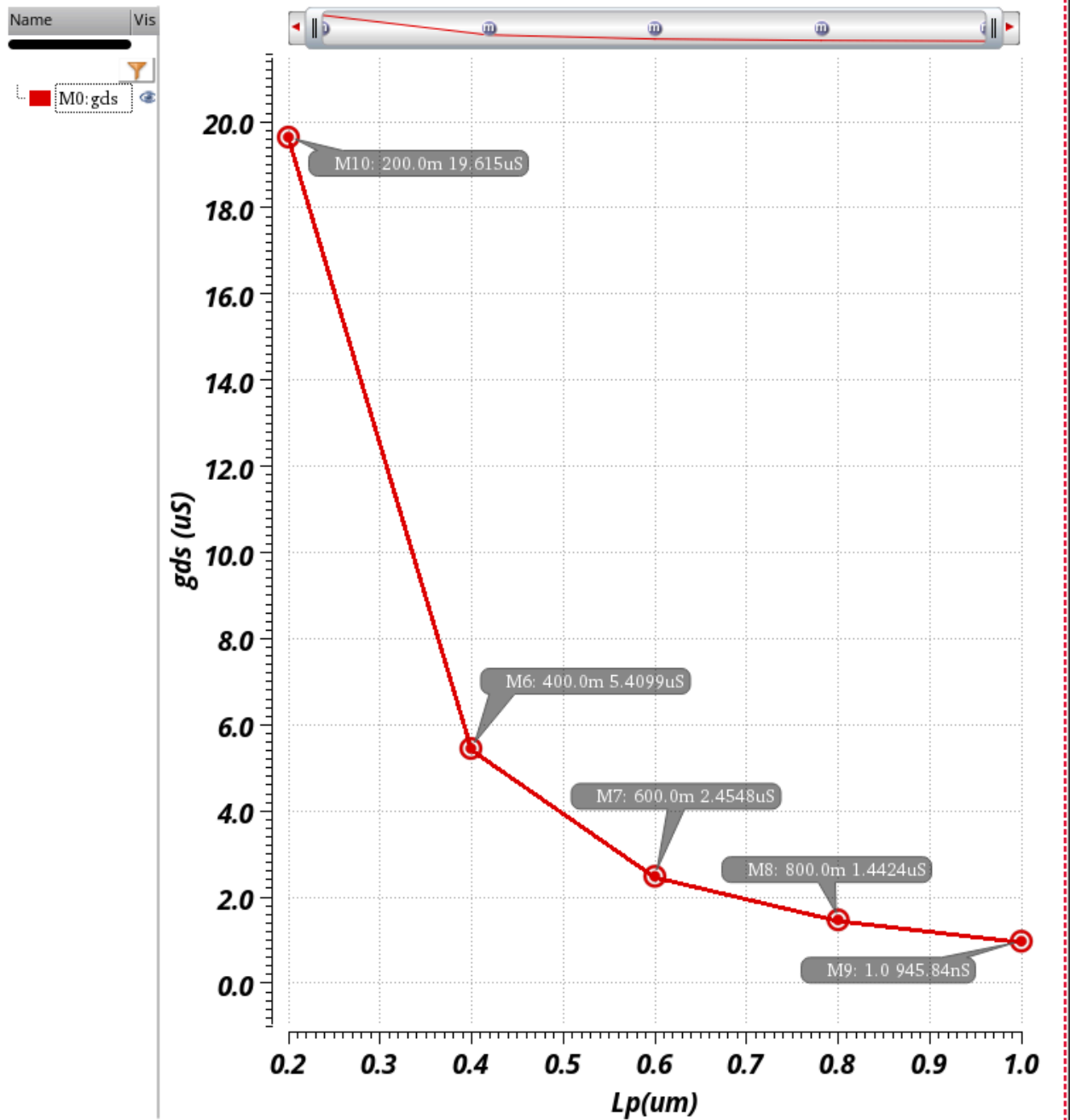


FIG:-3.0 DRAIN SOURCE CONDUCTANCES(g_{ds}) AS FUNCTION OF GATE LENGTH (PMOS)

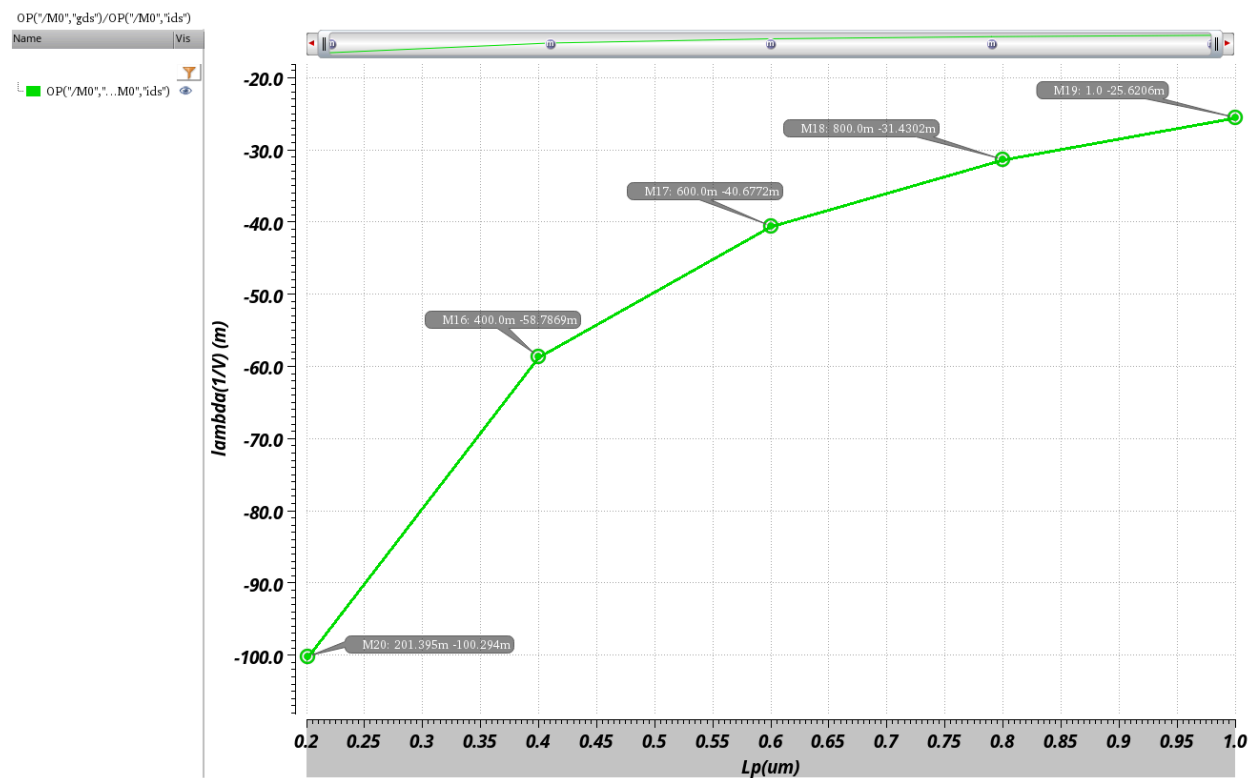


FIG:-3.1 LAMBDA AS FUNCTION OF GATE LENGTH (PMOS)

$L_p(\mu m)$	$\lambda(1/V)$
0.2	-100.2
0.4	-58.7
0.6	-40.677
0.8	-31.43
1	-25.62

As length increases channel length modulation decreases because length and λ are inversely proportional. We can observe the trend.

Lp(um)	Vtp(mV)	Isd(uA)	gm(uS)	gsd(uS)	lambda(1/V)
0.2	-452.22	-195.01	213.02	19.615	-100.2
0.4	-420.54	-92.02	105.08	5.4	-58.7
0.6	-414.8	-60.348	70.898	2.45	-40.677
0.8	-407.13	-45.81	54.668	1.44	-31.43
1	-401.4	-36.91	44.28	.945	-25.62

PMOS IN SATURATION

TASK 3

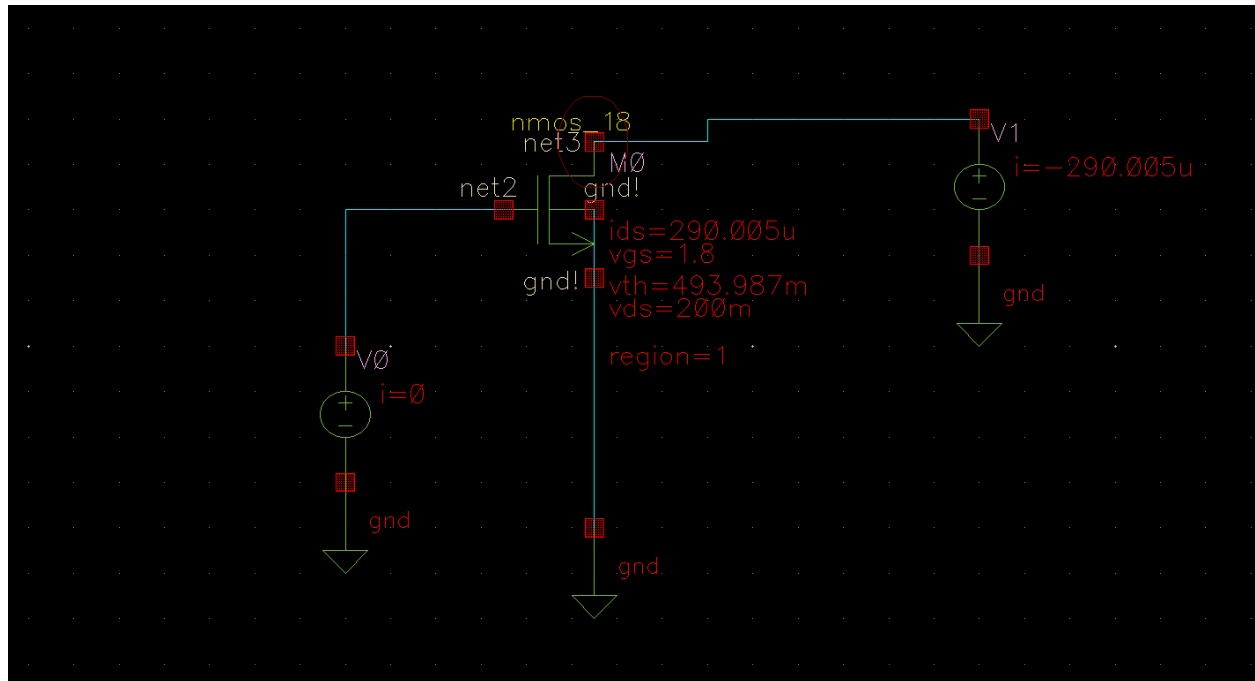


FIG:-3.0 SCHEMATIC CIRCUIT OF NMOS IN LINEAR REGION

PMOS

$$L_p = 0.2 \mu m ; W_p = 1.3 \mu m$$

β_{eff}	$W_n (\mu m)$
174.6×10^6	0.5 μ
232.9×10^6	0.7
291.2×10^6	0.9
349.5×10^6	1.1
428×10^6	1.3

$$\beta_{eff} = \mu_{n/ox} \left(\frac{W}{L} \right)$$

$$\mu_{n/ox} = \frac{\beta_{eff}}{(W/L)} = \frac{349.5 \times 10^6}{\left(\frac{1.1 \times 10^6}{0.2 \times 10^6} \right)} = \frac{349 \times 10^6}{5.5}$$

$$\mu_{n/ox} = 63.45 \mu A/V^2$$

NMOS

$$W_n = 0.9 \mu m ; L_n = 0.2 \mu m$$

$$\beta = \mu_{n/ox} \left(\frac{W}{L} \right)$$

$$\Rightarrow \frac{1.35 \times 10^3}{\left(\frac{0.9}{0.2} \right)} = \mu_{n/ox} \Rightarrow 0.3 m A/V^2$$

Calculation of $\mu_{n/ox}$ for PMOS and NMOS transistor

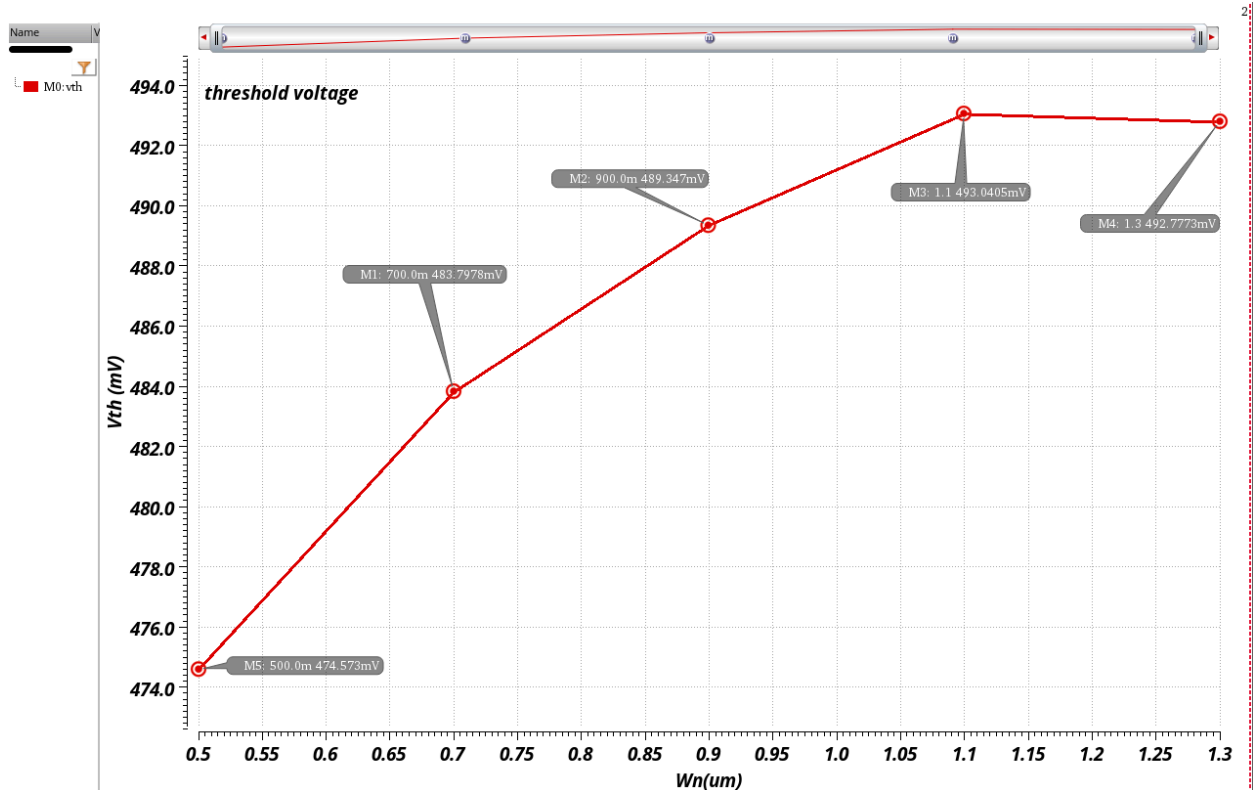


FIG:-3.1 THESHOLD VOLTAGE AS FUNCTION OF WIDTH (NMOS) LINEAR REGION

The threshold voltage (V_{th}) remains nearly constant as the width increases from 500 nm to 1300 nm for both NMOS and PMOS

V_{th} depends mainly on the gate length and doping, not width, so it doesn't change much.

NMOS will show a positive V_{th} (0.4-0.7V), while PMOS will show a negative V_{tp} (-0.4 to -0.7V)

@0.5(w_n)=474.573mV

@1.3(w_n)=492.77mv

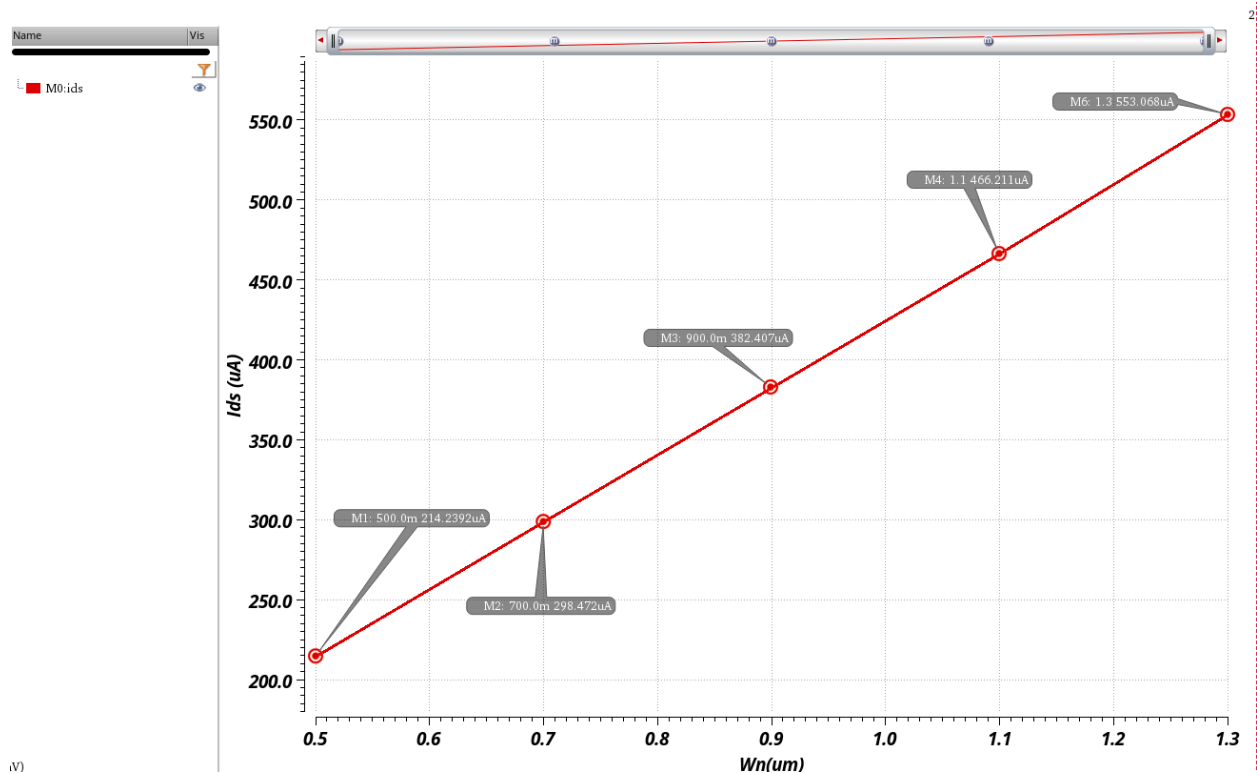


FIG:-3..2 DRAIN SOURCE CURRENTS FUNCTION AOF WIDTH (NMOS) LINEAR REGION

$$I_D = 0, \text{ for } V_{GS} < V_T$$

$$I_D(\text{lin}) = \frac{\mu_n \cdot C_{ox}}{2} \cdot \frac{W}{L} \cdot [2 \cdot (V_{GS} - V_T) V_{DS} - V_{DS}^2] \quad \text{for } V_{GS} \geq V_T$$

and $V_{DS} < V_{GS} - V_T$

The drain current (I_{ds}) increases steadily as the width increases from 500 nm to 1300 nm for both NMOS and PMOS in both linear and saturation regions. This increase is more pronounced in the saturation region due to the quadratic relationship between I_{ds} and width in saturation ($I_{ds} \propto (W/L) \cdot (V_{gs} - V_{th})^2$). A wider channel provides a larger effective area for current conduction, reducing resistance and allowing more charge carriers (electrons for NMOS, holes for PMOS) to flow. In simulation, I saw a near-linear rise in I_{ds} with width, with NMOS exhibiting higher current magnitudes than PMOS due to the higher mobility of electrons compared to holes. The difference between linear and saturation regions would show a steeper slope in saturation, reflecting the stronger width dependence.

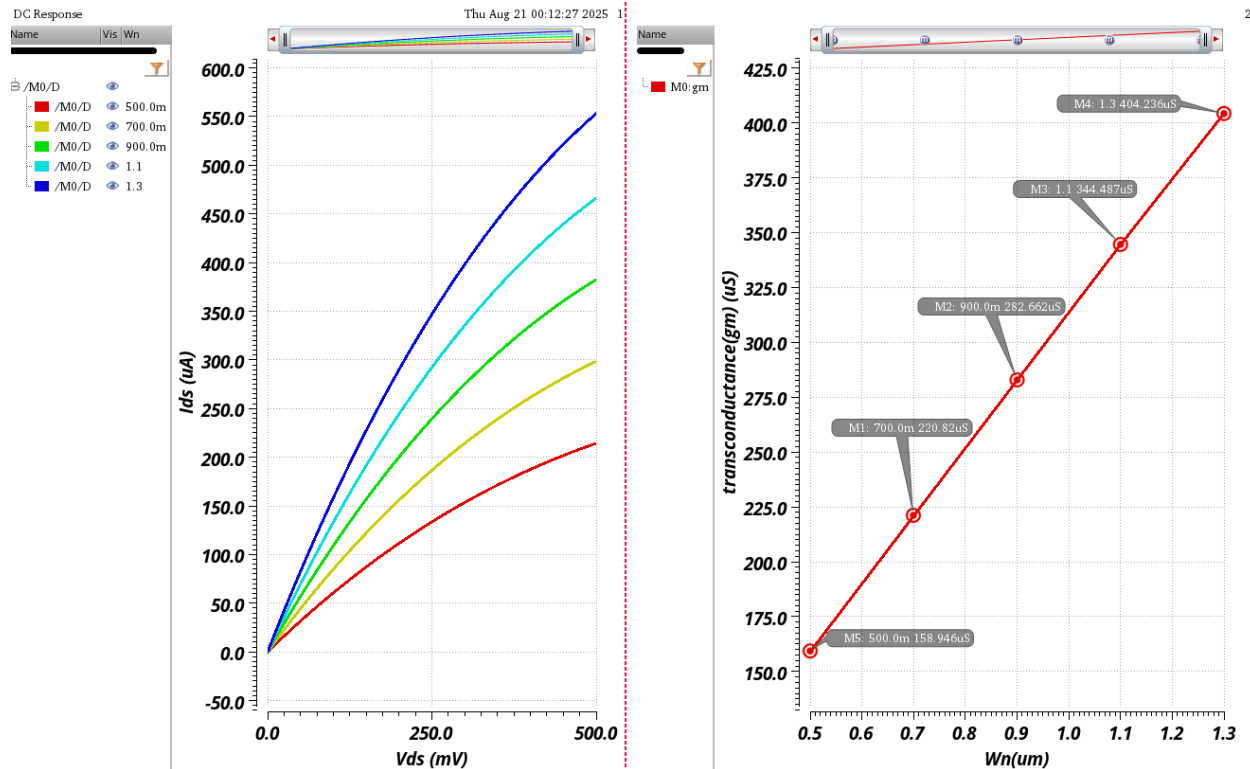


FIG:-3.3 TRANSCONDUCTANCE FUNCTION OF WIDTH (NMOS) LINEAR REGION

The transconductance (g_m), which measures how effectively the transistor controls I_{ds} with changes in gate-source voltage (V_{gs}), increases as the width increases from 500 nm to 1300 nm for both NMOS and PMOS in both regions, with a more noticeable rise in the saturation region. Mathematically, g_m is proportional to width ($g_m \propto W * \mu * C_{ox} * (V_{GS} - V_{th}) / L$), so a larger W directly enhances g_m . Plots likely showed a consistent upward trend, with NMOS displaying higher g_m values than PMOS due to electron mobility advantages. The increase is less steep in the linear region, where g_m depends linearly on I_{ds} .

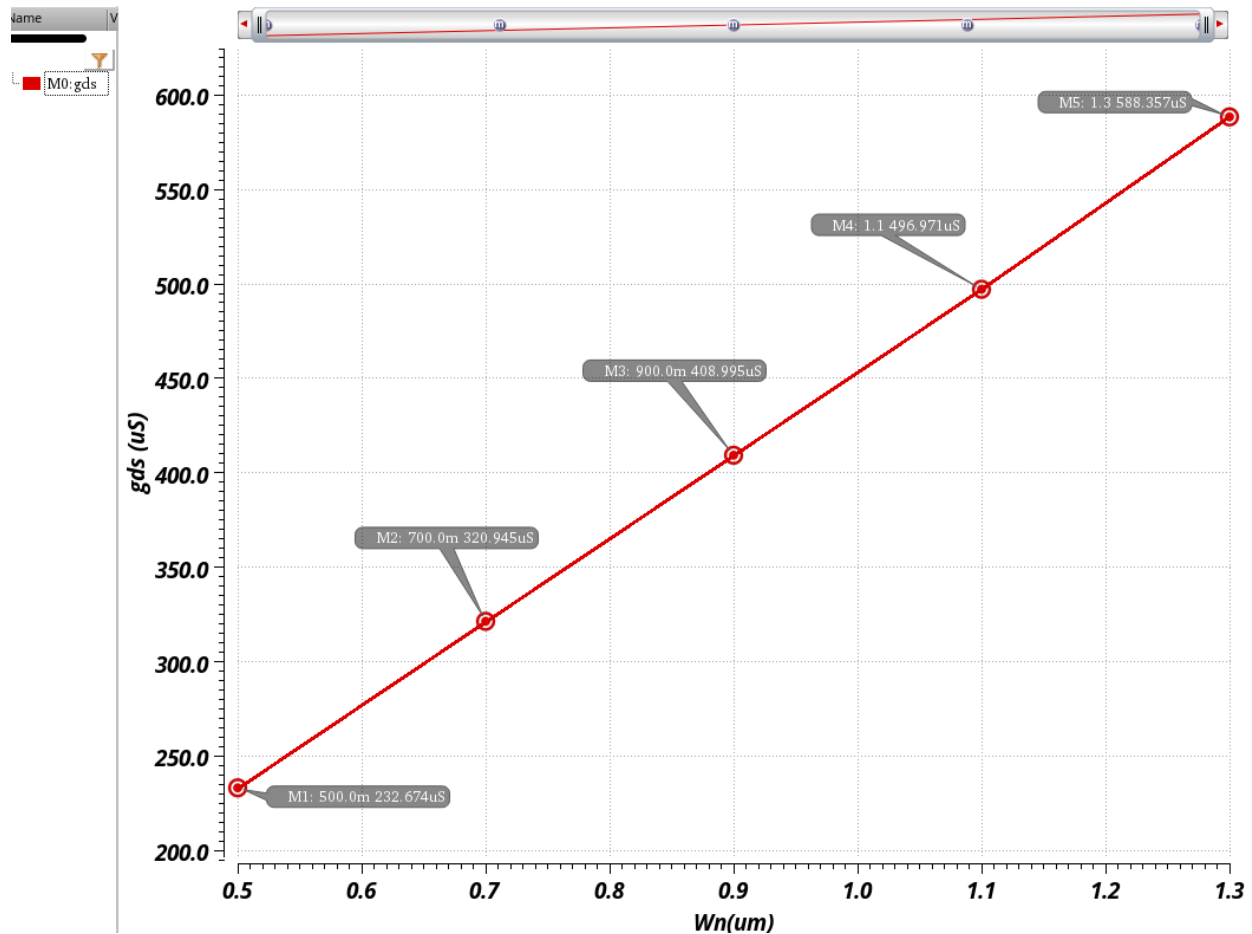


FIG:- 3.4 DRAIN SOURCE CONDUCTANCE FUNCTION OF WIDTH (NMOS) LINEAR REGION

The output conductance (g_{ds}), which indicates how much I_{ds} changes with drain-source voltage (V_{ds}), remains relatively constant or shows a slight increase as the width increases from 500 nm to 1300 nm for both NMOS and PMOS in both linear and saturation regions. g_{ds} is primarily influenced by channel-length modulation and early voltage effects, which depend more on L_G and V_{DS} than on width. However, a wider channel might slightly enhance leakage or modulation effects, leading to a minor upward trend. In simulation, the g_{ds} curve showed a gentle slope, with similar behavior for NMOS and PMOS.

Nmos in linear region

Wn(um)	Vtn(mV)	Ids(mA)	gm(uS)	gds(uS)
0.5	474.573	214.239	158.946	232.674

0.7	483.79	298.472	220.82	320.945
0.9	489.347	382.4	282.66	408.995
1.1	493.04	466.211	344.48	496.971
1.3	492.77	553.06	404.23	588.357

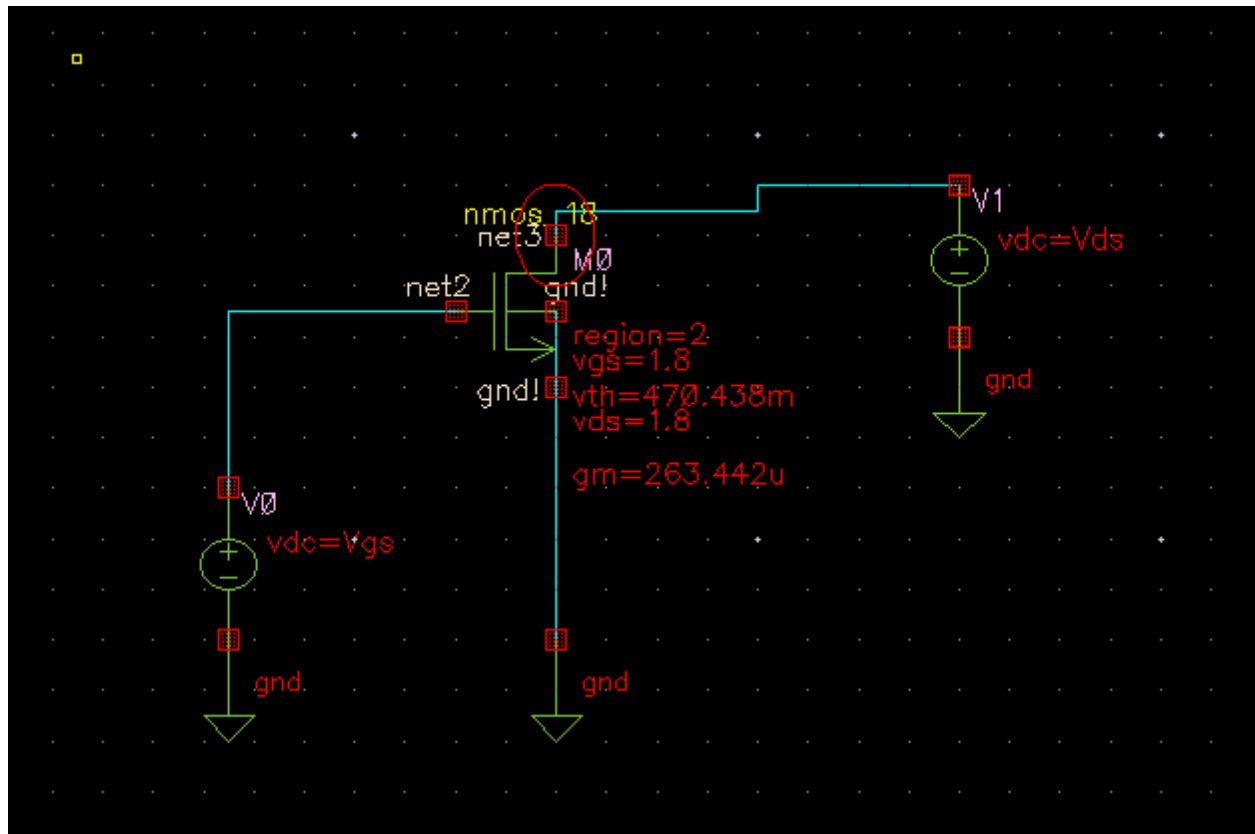


FIG:-3.5 SHEMATIC CIRCUIT FOR NMOS IN SATURATION

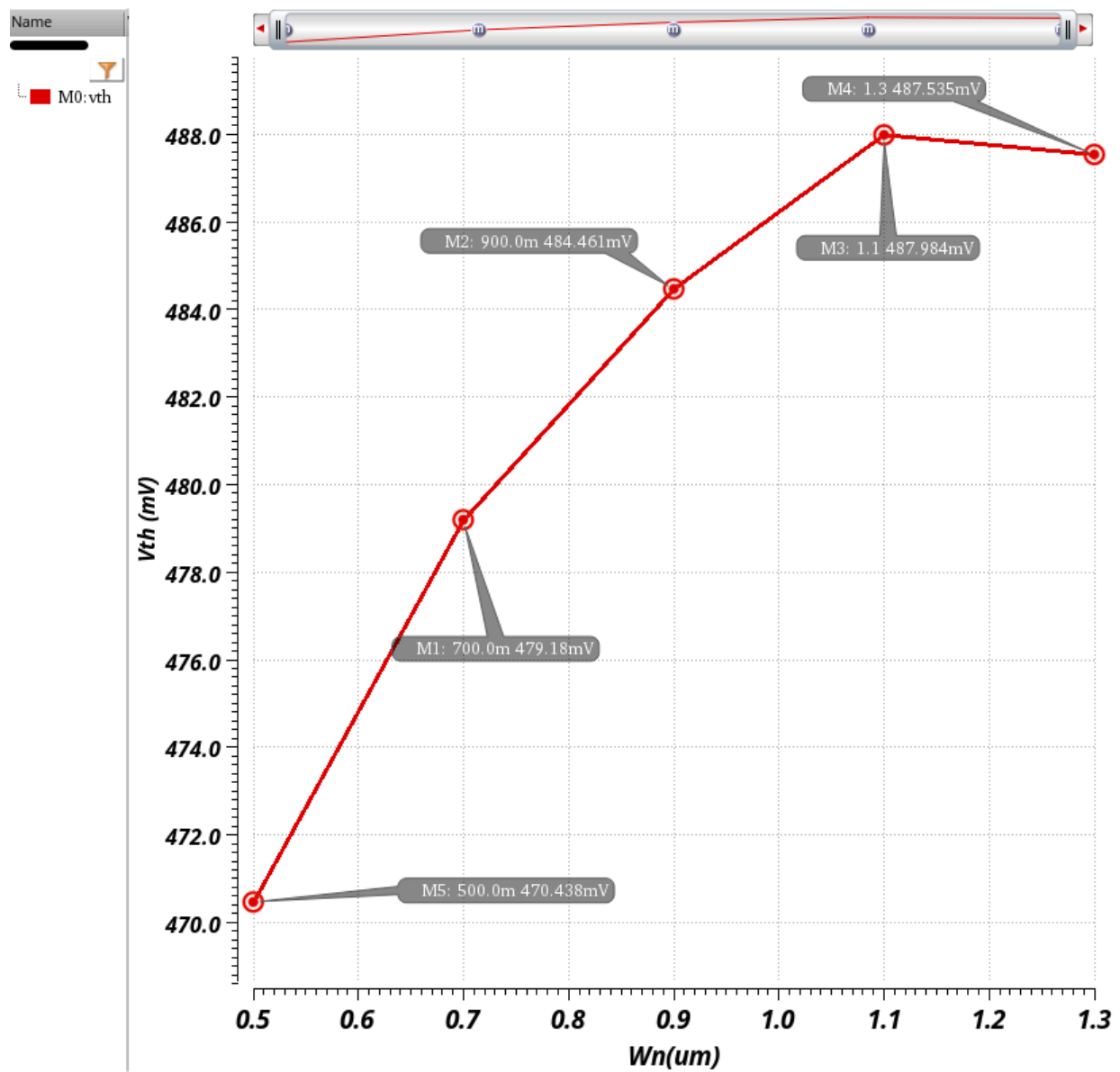


FIG:-3.6 THRESHOLD VOLTAGE AS FUNCTION OF WIDTH (NMOS) SATURATION REGION

The threshold voltage (V_{th}) remains nearly constant with increasing width in both linear and saturation regions

VTH(mV)	Wn(um)
470.438	0.5
479.18	.7
484.64	.9

Name

M0:ids

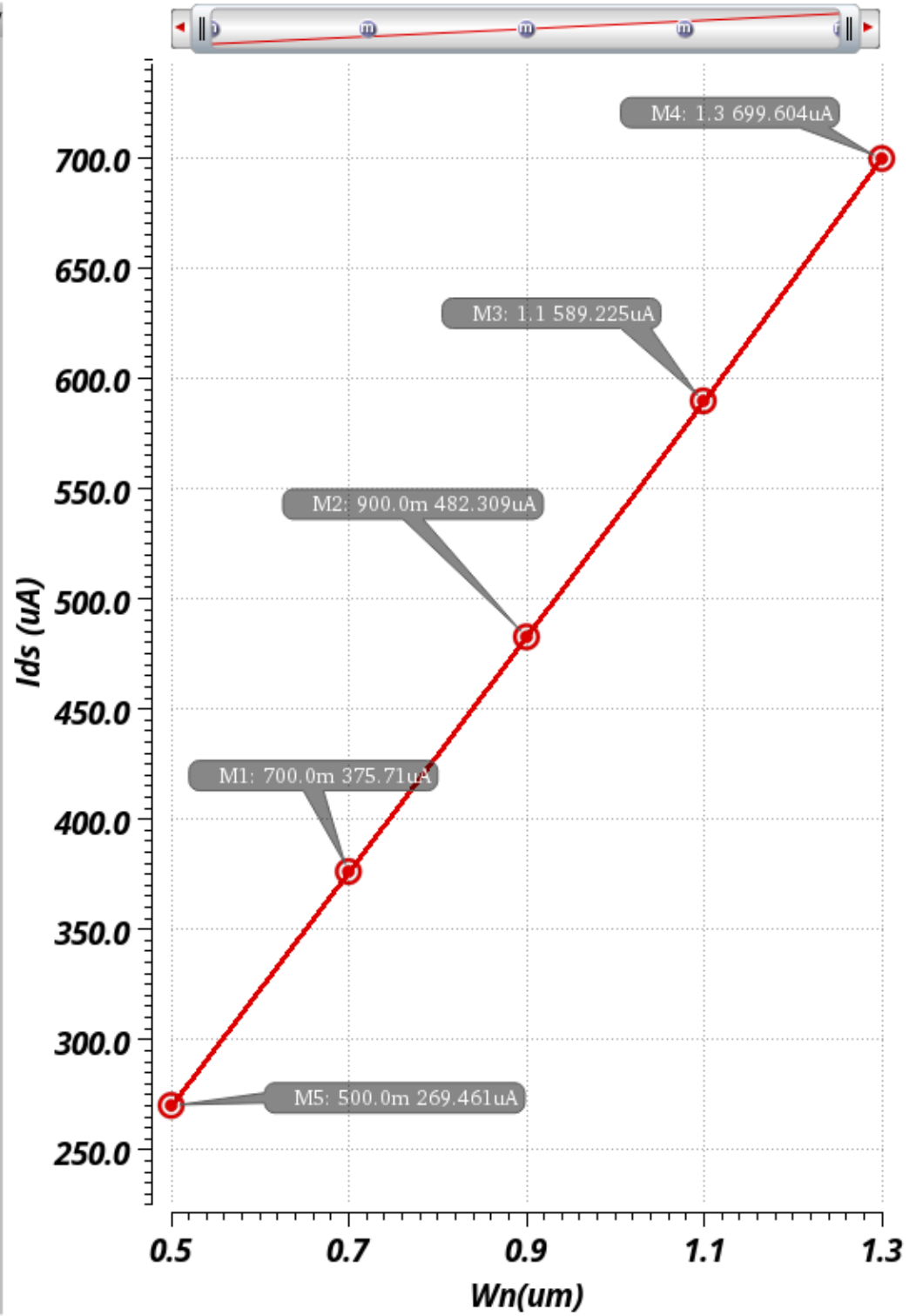


FIG:- 3.7 DRAIN SOURCE CURRENTS FUNCTION OF WIDTH (NMOS) SATURATION REGION

Wn(um)	Ids(uA)
0.5	269.461
0.7	375.71
0.9	482.309
1.1	589.225
1.3	699.604

Drain to source current increases as increases in the width of the transistor because current is directly proportional to width.

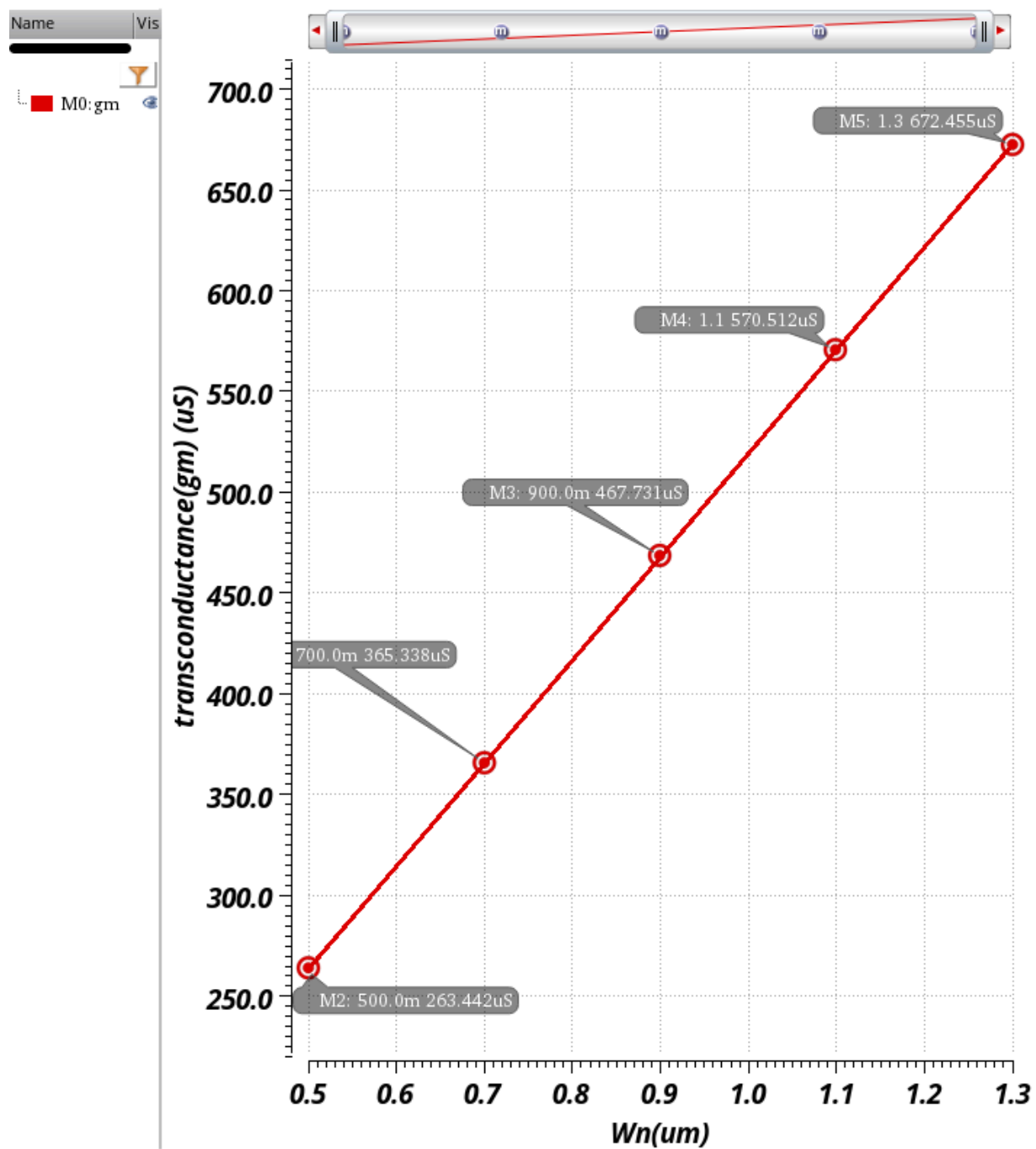


FIG:-3.8 TRANSCONDUCTANCE FUNCTION OF WIDTH (NMOS) SATURATION REGION

In saturation, where $g_m = 2 \cdot I_{ds} / (V_{gs} - V_{th})$,

Wn(um)	gm(uS)
0.5	263.442
0.7	365.338
0.9	467.7
1.1	570.512
1.3	672.455

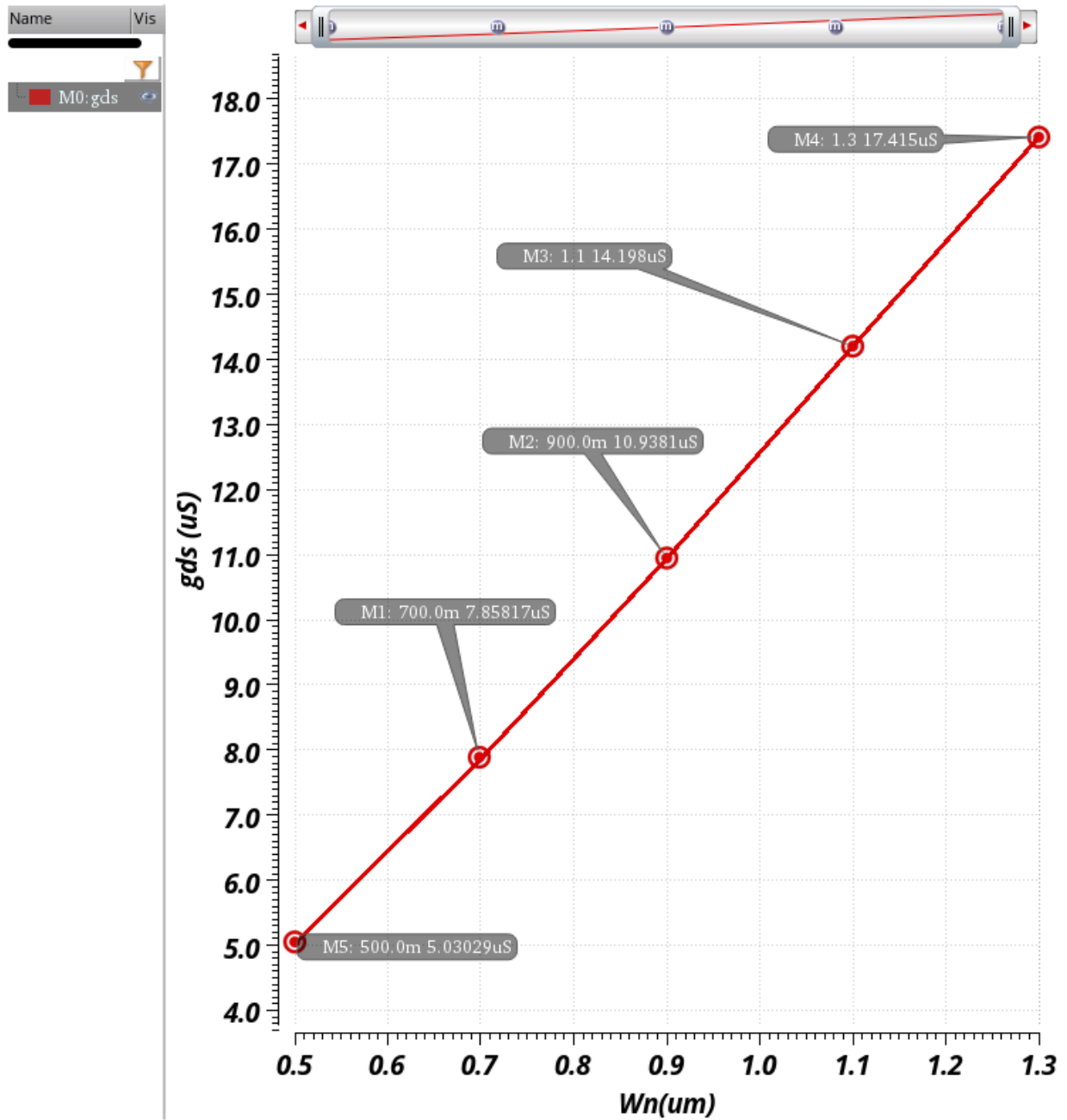


FIG:-3.9 DRAIN SOURCE CONDUCTANCE FUNCTION OF WIDTH (NMOS) SATURATION REGION

$W_n(\text{um})$	$g_{ds}(\text{uS})$
0.5	5.03
0.7	7.8

0.9	10.93
1.1	14.198
1.3	17.4

Wn(um)	Vtn(mV)	Ids(uA)	gm(uS)	gds(uS)
0.5	470.438	269.461	263.442	5.03
0.7	479.18	375.71	365.338	7.8
0.9	484.64	482.309	467.7	10.93
1.1	487.984	589.225	570.512	14.198
1.3	487.535	699.604	672.455	17.4

Nmos in saturation

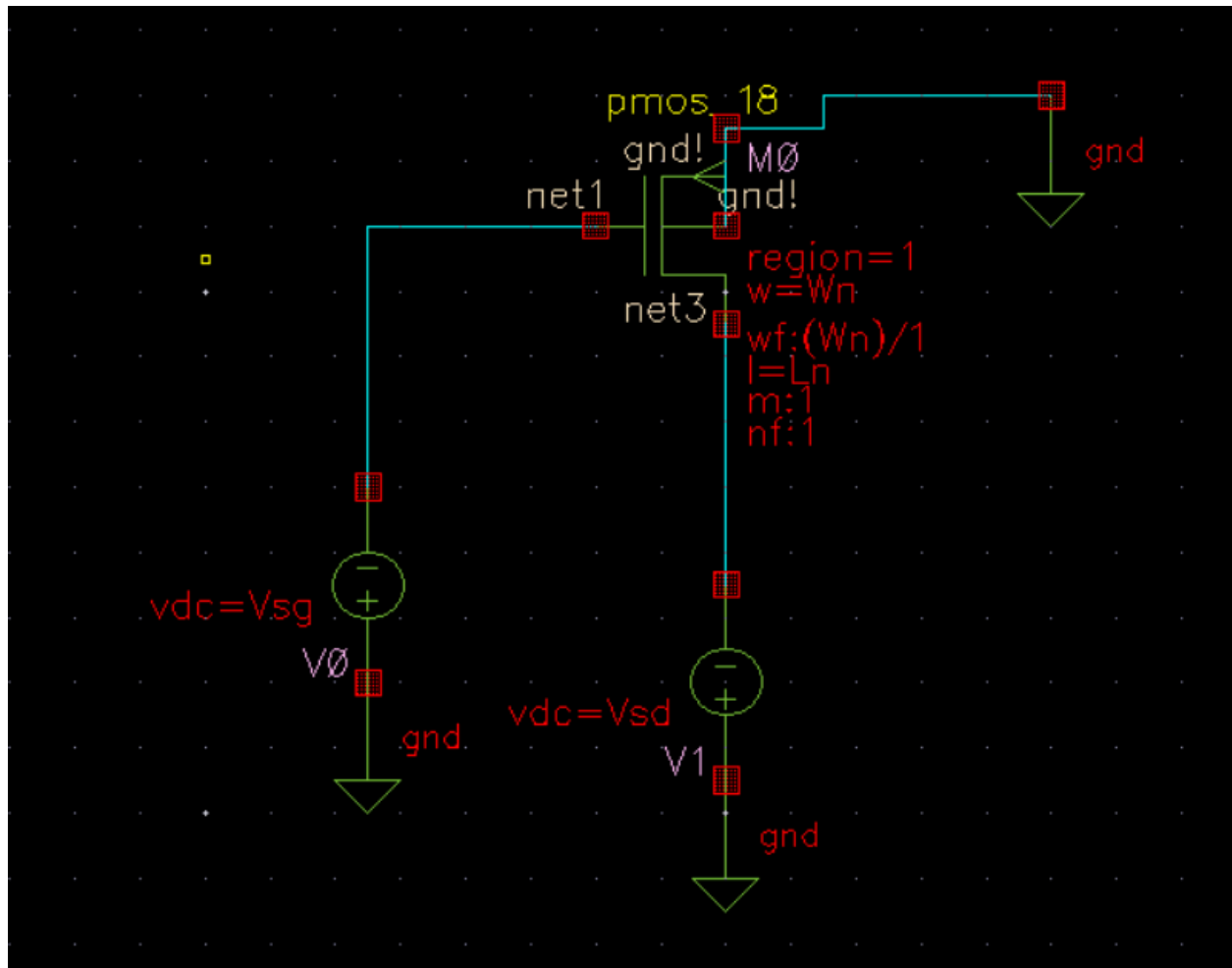


FIG:- 4.0 SCHEMATIC CIRCUIT OF PMOS IN LINEAR REGION

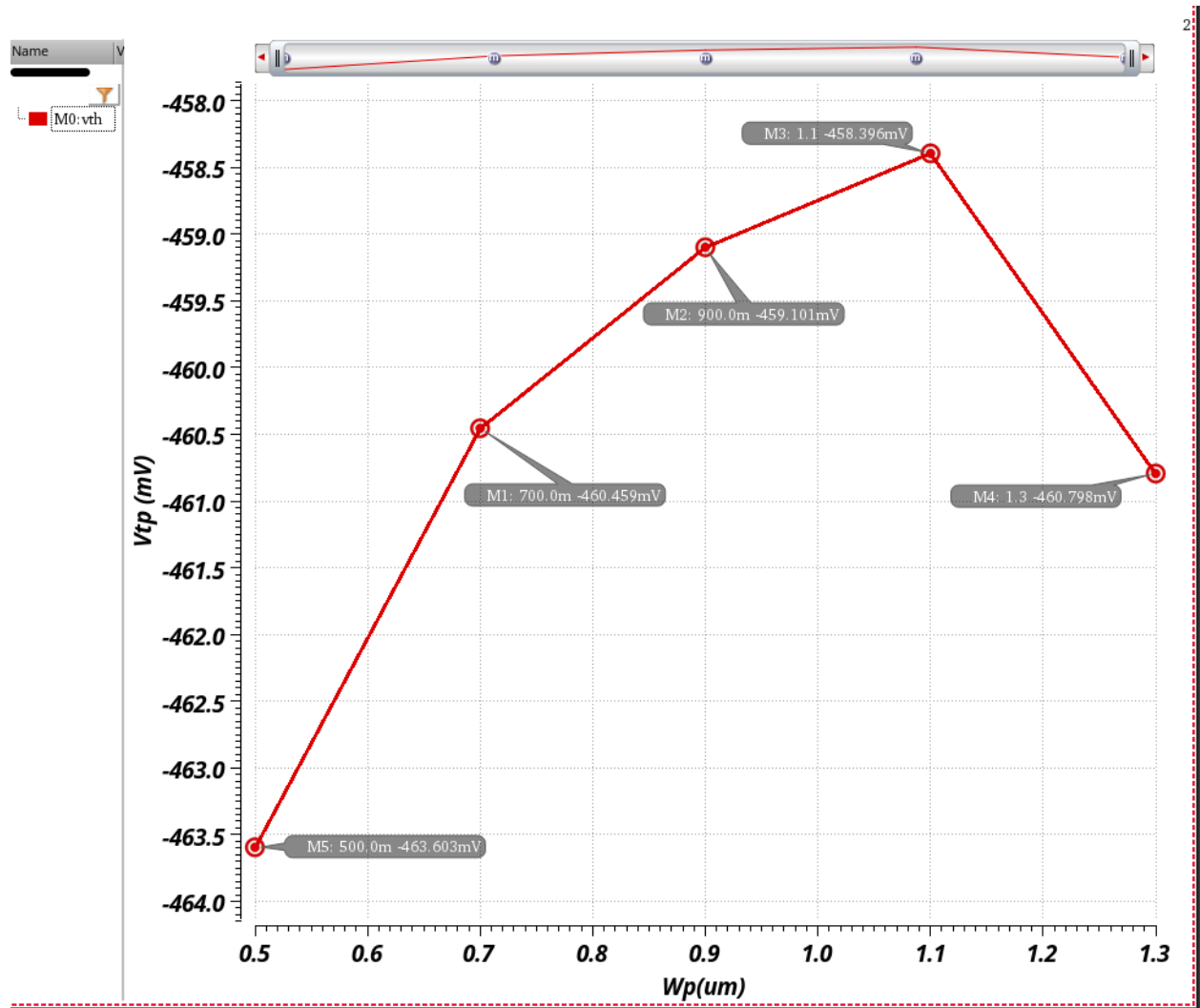


FIG:- 4.1 THRESHOLD VOLTAGE AS FUNCTION OF WIDTH (PMOS) LINEAR REGION

Wp(um)	Vtp(mV)
0.5	-463.6
0.7	-460.4
0.9	-459.1
1.1	-458.396
1.3	-460.7

THRESHOLD VOLTAGE IS APPROXIMATELY SAME BECAUSE VTP IS INDEPENDENT OF WIDTH

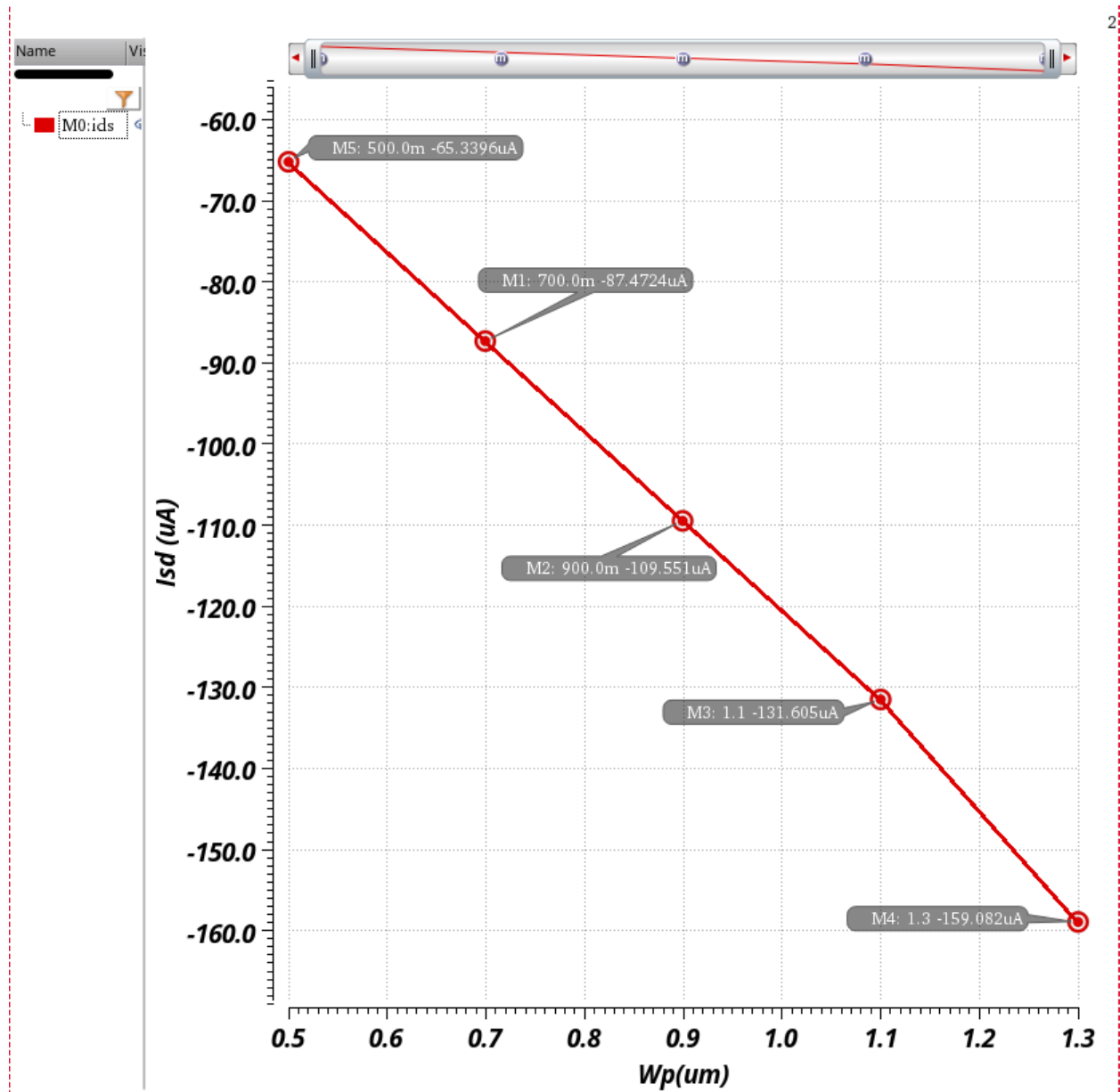


FIG:- 4.2 SOURCE DRAIN CURRENT AS FUNCTION OF WIDTH (PMOS) LINEAR REGION

Wp(um)	I _{sd} (uA)
0.5	-65.33
0.7	-87.472
0.9	-109.55
1.1	-131.605
1.3	-159.08

As width increases the current also increases because its directly proportional to width

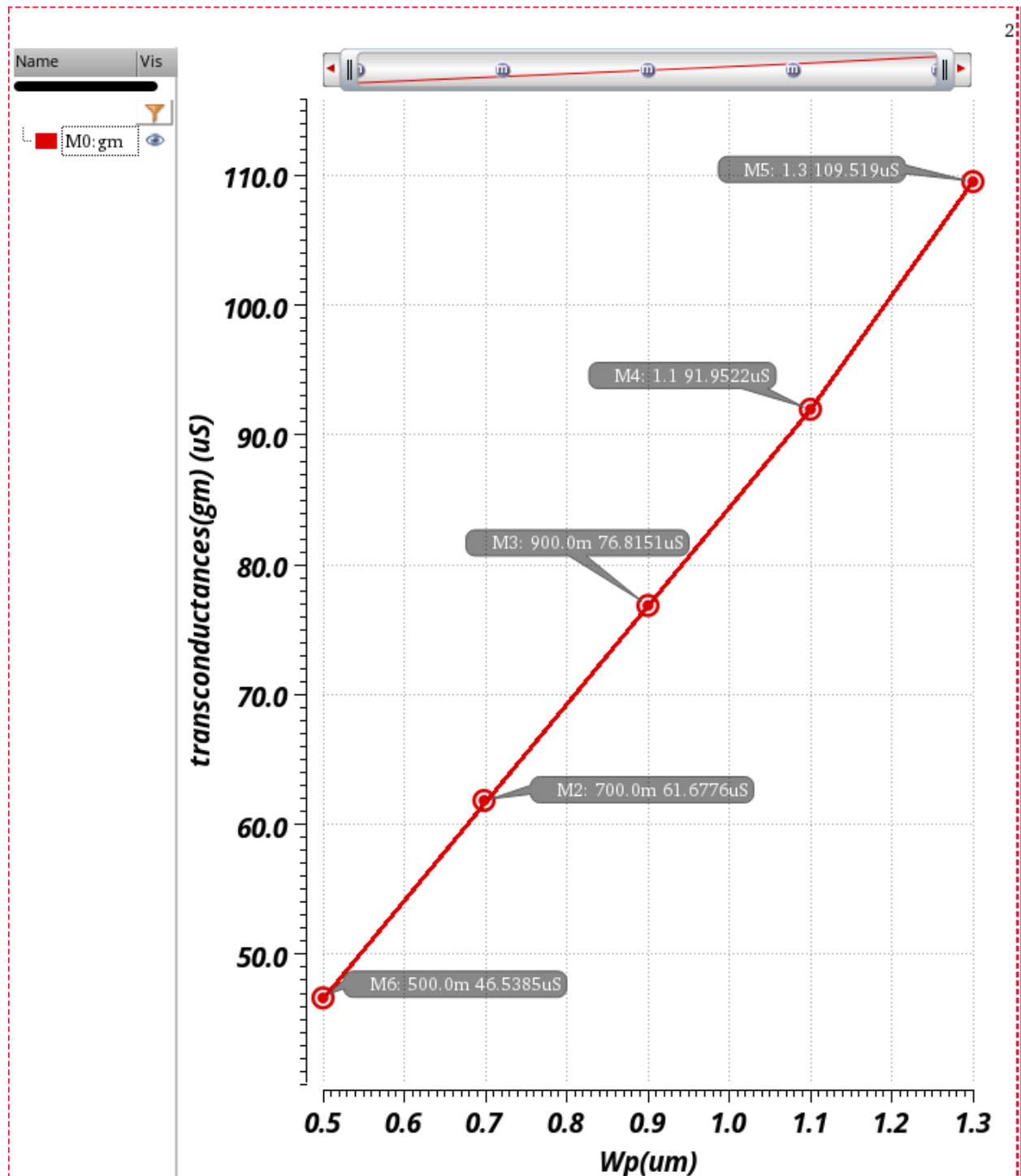


FIG:- 4.3 TANSCONDUCTANCE AS FUNCTION OF WIDTH (PMOS) LINEAR REGION

W_p (um)	g_m (uS)
0.5	46.538

0.7	61.677
0.9	76.8151
1.1	91.95
1.3	109.519

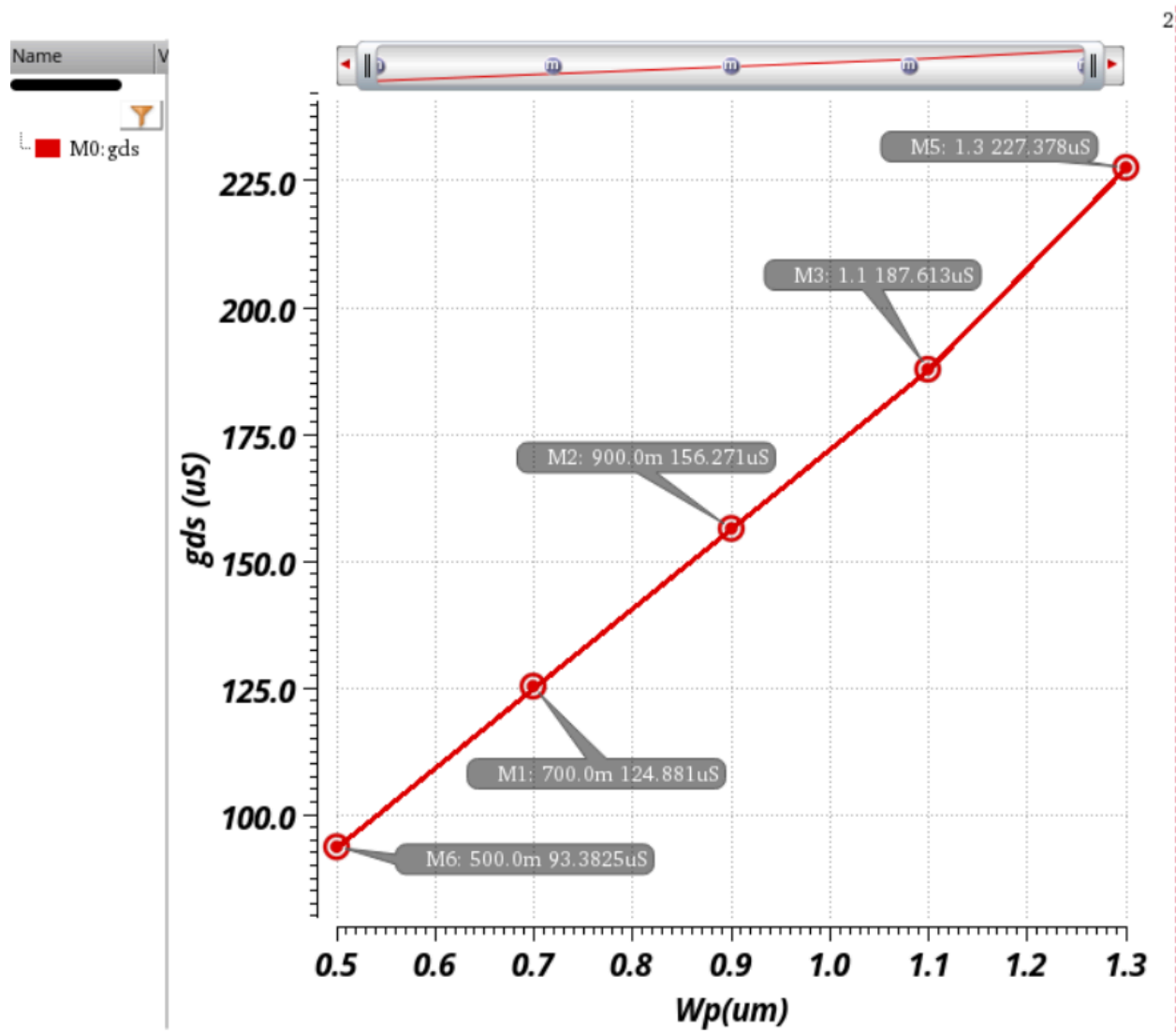


FIG:- 4.4 CONDUCTANCE AS FUNCTION OF WIDTH (PMOS) LINEAR REGION

Wp(um)	gsd(uS)
0.5	93.3825
0.7	124.881

0.9	156.271
1.1	187.61
1.3	227.378

PMOS in linear region

Wp(um)	Vtp(mV)	Isd(uA)	gm(uS)	gsd(uS)
0.5	-463.6	-65.33	46.538	93.3825
0.7	-460.4	-87.472	61.677	124.881
0.9	-459.1	-109.55	76.8151	156.271
1.1	-458.396	-131.605	91.95	187.61
1.3	-460.7	-159.08	109.519	227.378

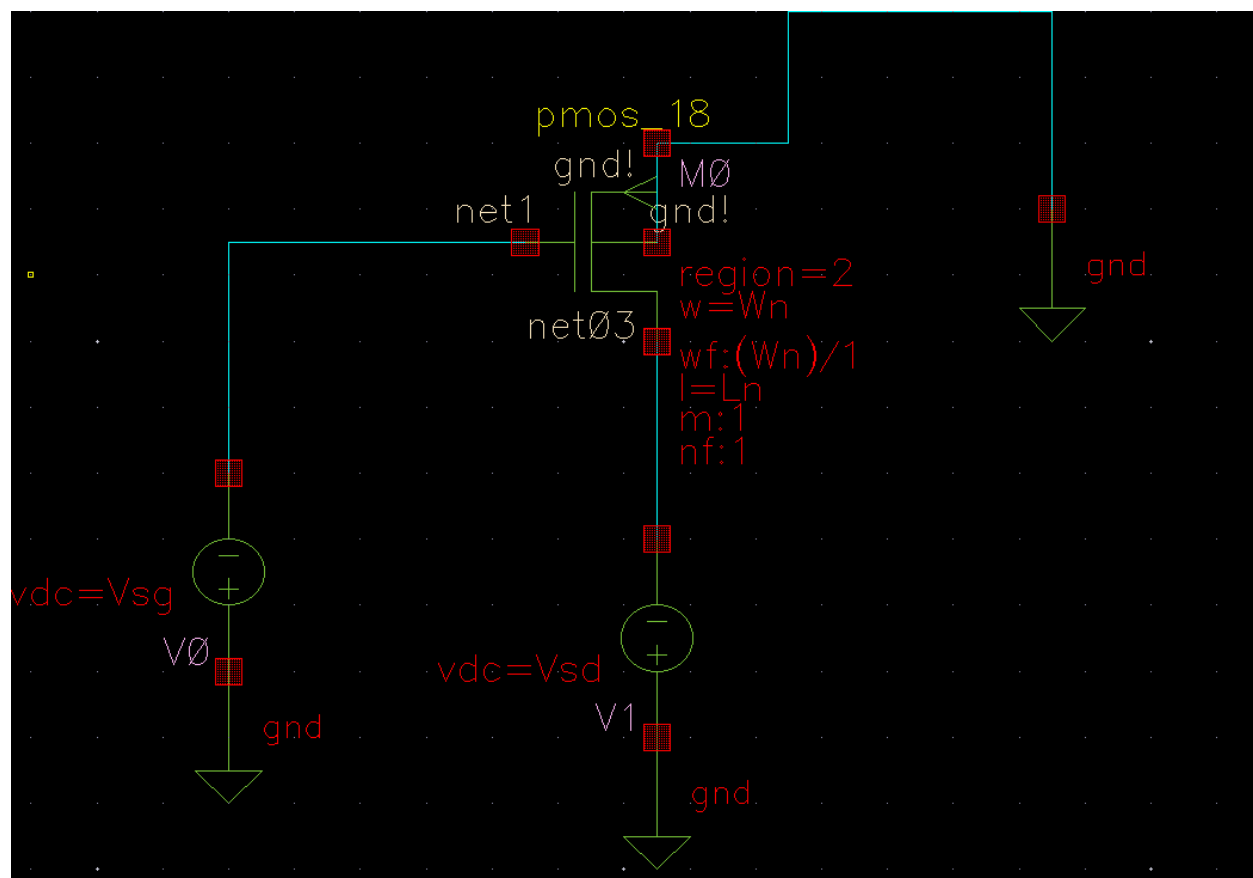


FIG :- 4.5 SHEMATIC CIRCUIT OF PMOS IN SATURATION REGION

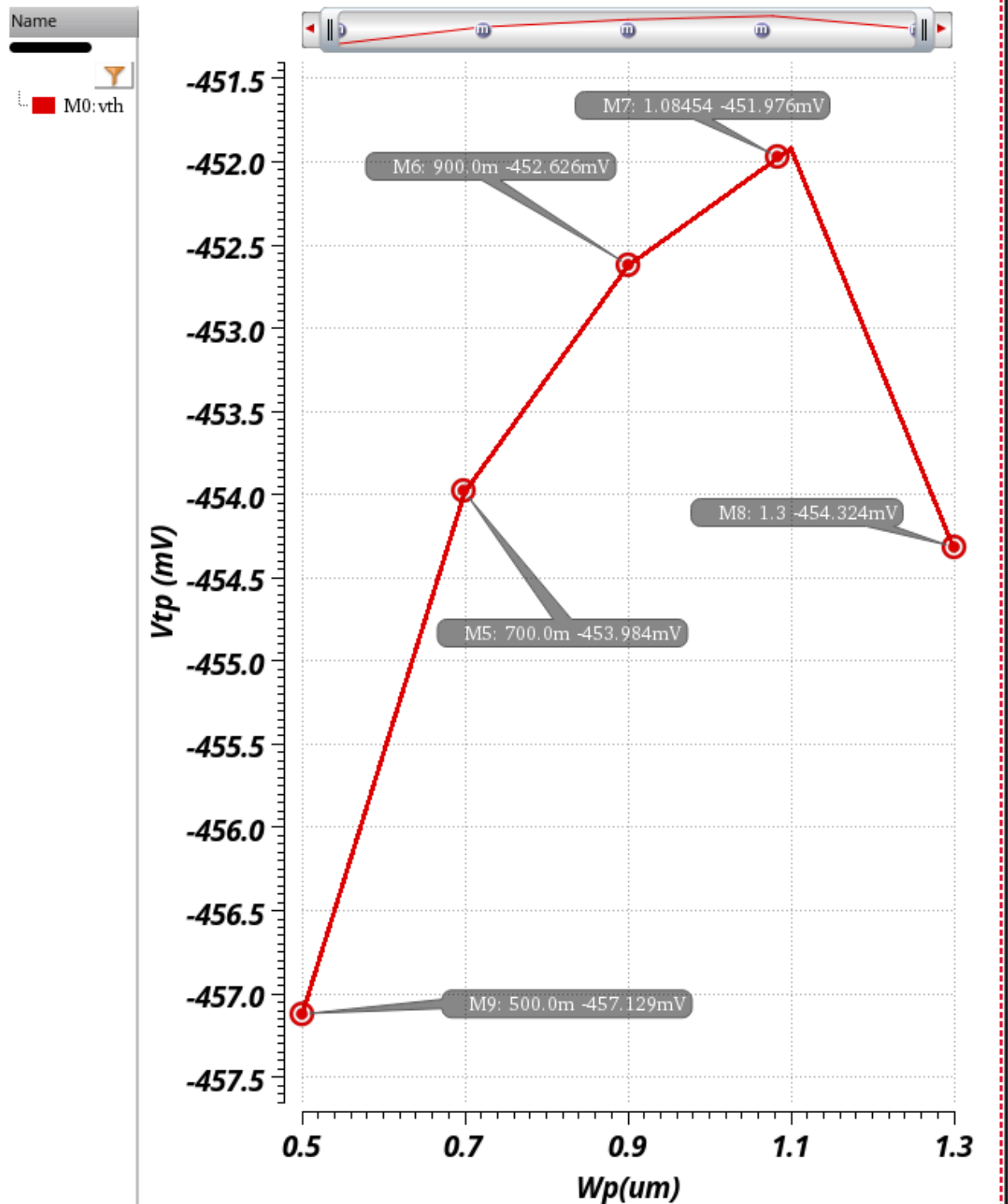


FIG:- 4.6 THRESHOLD VOLTAGE FUNCTION OF WIDTH (PMOS) SATURATION REGION

$W_p(\mu m)$	$V_{tp}(mV)$
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0.5	-457.129
0.7	-453.9
0.9	-452.62
1.1	-451.9
1.3	-454.324

Threshold voltage is approximately constant because it doesnt depend upon the width of the transistor only depends upon the length.

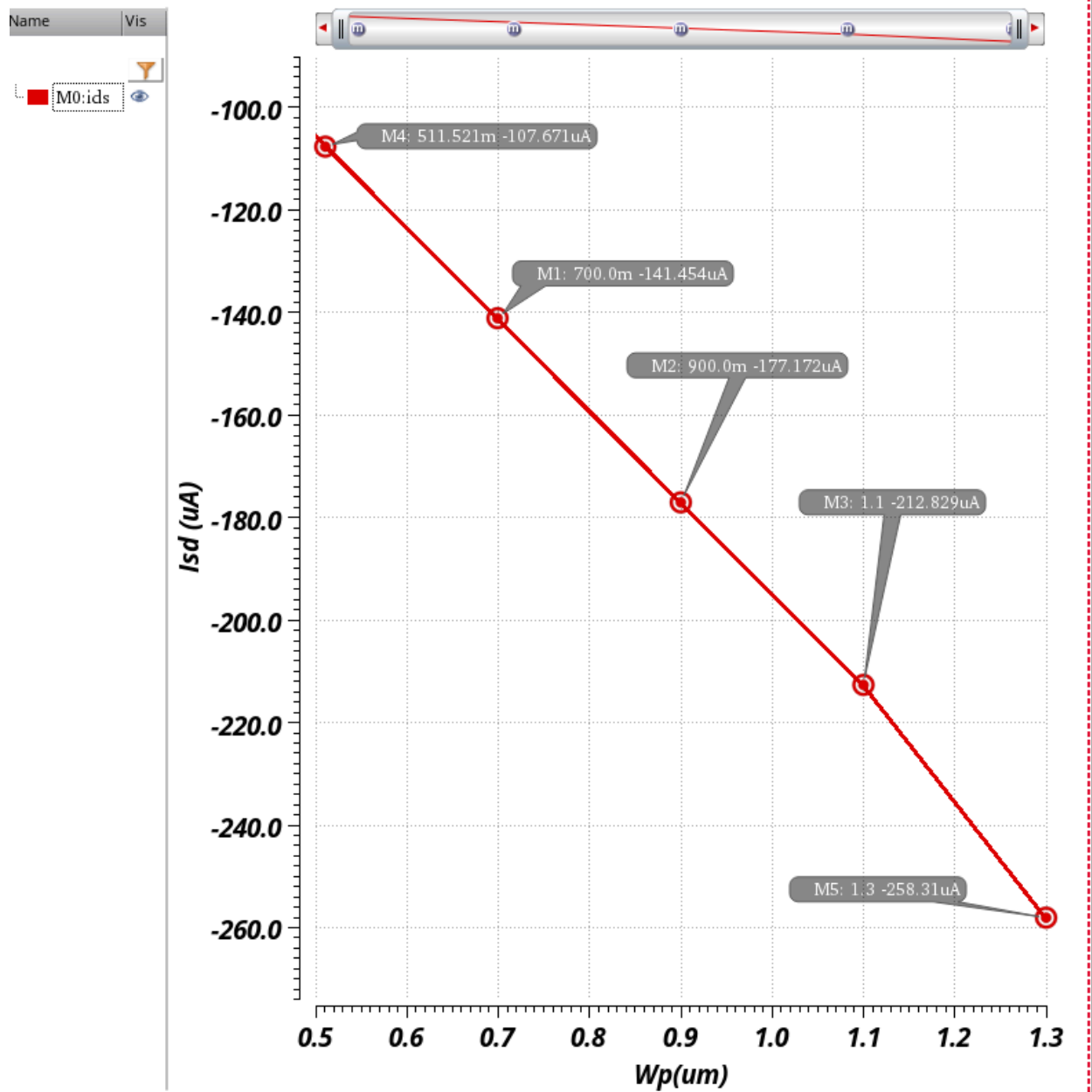


FIG:- 4.7 SOURCE DRAIN CURRENT FUNCTION OF WIDTH (PMOS) SATURATION REGION

$W_p(\mu m)$	$I_{sd}(\mu A)$
0.5	-107.671
0.7	-141.454
0.9	-177.172

1.1	-212.8
1.3	-258.3

Current increases as width increases i have noticed the dependence of current on width of the transistor

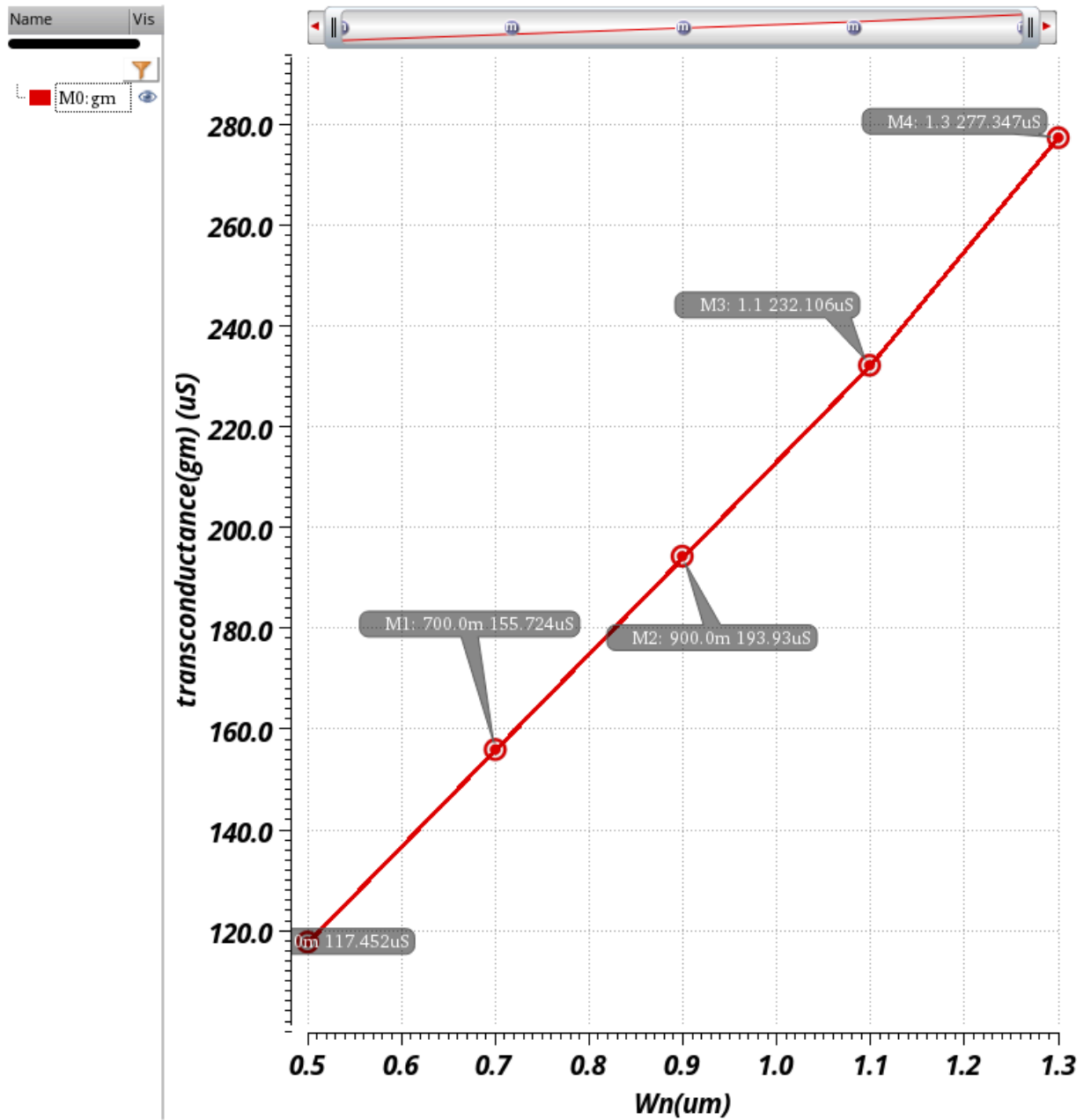


FIG:- 4.8 TRANSCONDUCTANCE FUNCTION OF WIDTH (PMOS) SATURATION REGION

$W_p(\mu\text{m})$	$g_m(\mu\text{S})$
0.5	117.452
0.7	155.724
0.9	193.93
1.1	232.1

As width increases the transconductance also increases because its depend on the width of the transistor

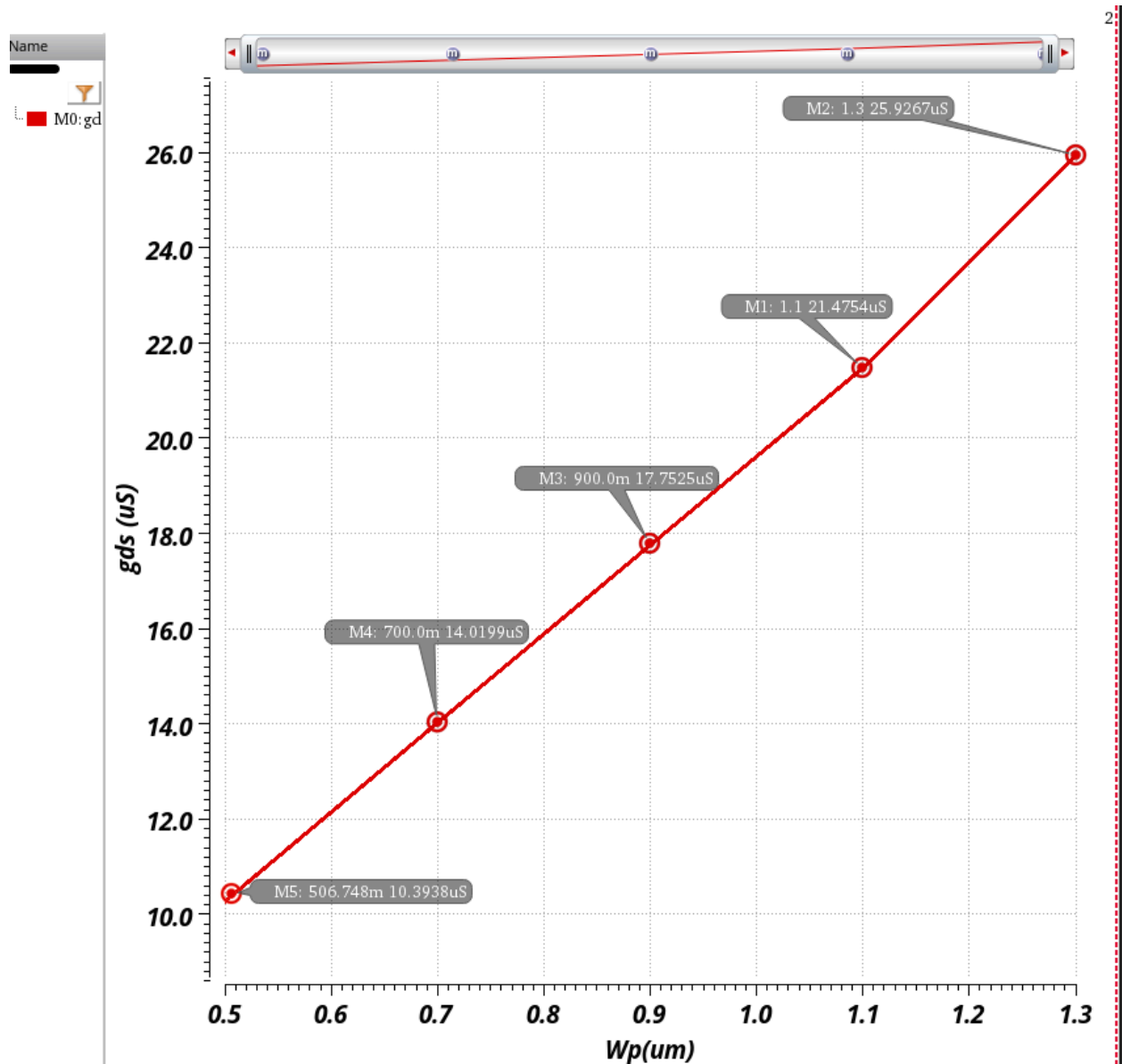


FIG:- 4.9 CONDUCTANCE(g_{ds}) FUNCTION OF WIDTH (PMOS) SATURATION REGION

$W_p(\mu m)$	$g_{ds}(\mu m)$
0.5	10.393
0.7	14.0199

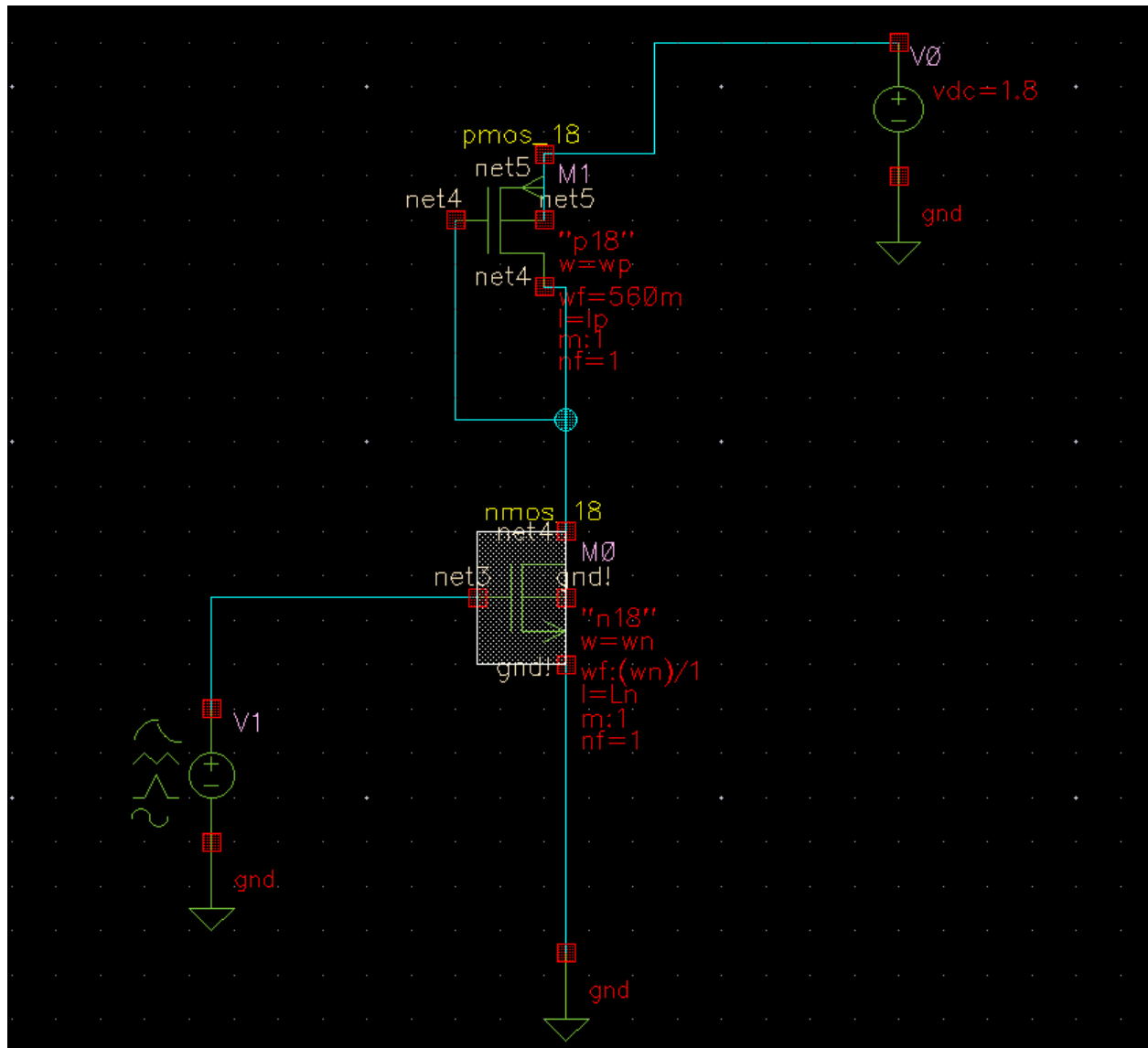
0.9	17.75
1.1	21.4
1.3	25.9

Here in this plot i have noticed that their is dependence of output conductance on width of the transistor

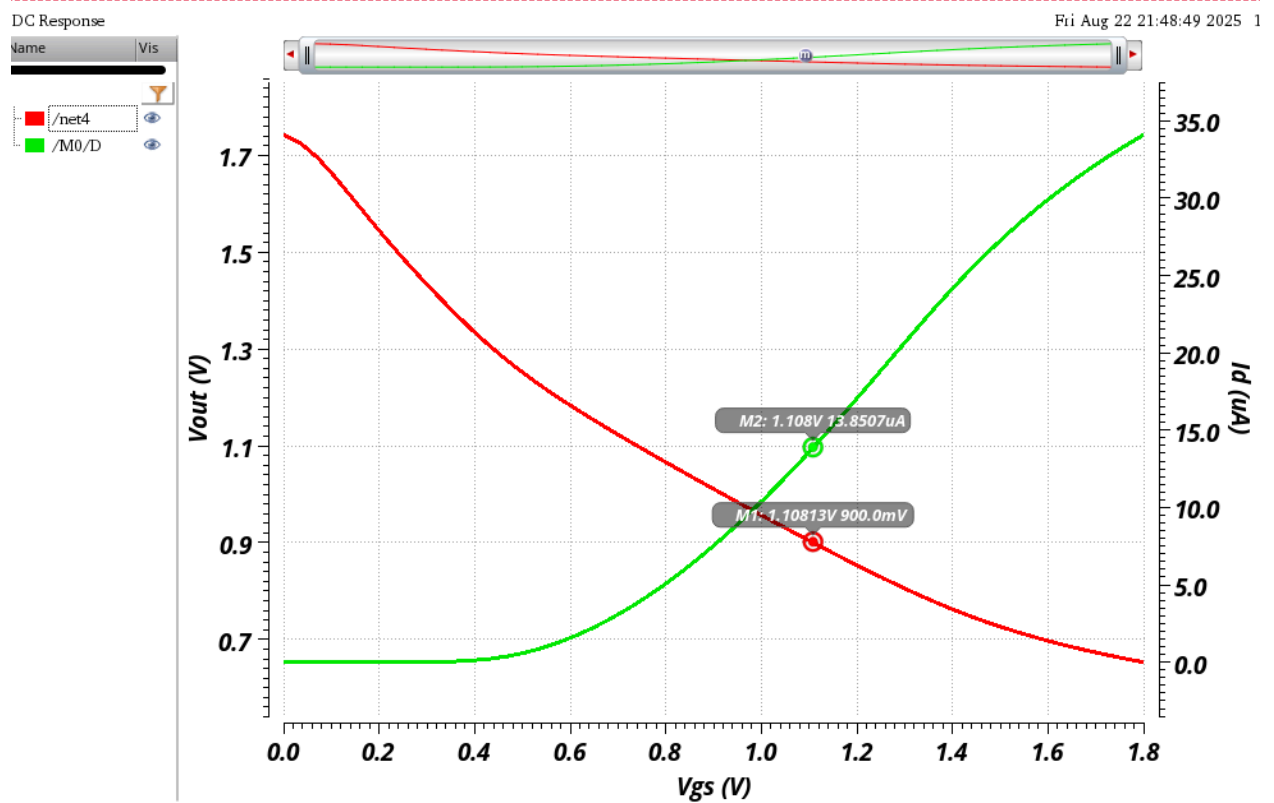
Pmos in Saturation

Wp(um)	Vtp(mV)	Isd(uA)	gm(uS)	gsd(uS)
0.5	-457.129	-107.671	117.452	10.393
0.7	-453.9	-141.454	155.724	14.0199
0.9	-452.62	-177.172	193.93	17.75
1.1	-451.9	-212.8	232.1	21.4
1.3	-454.324	-258.3	277.3	25.9

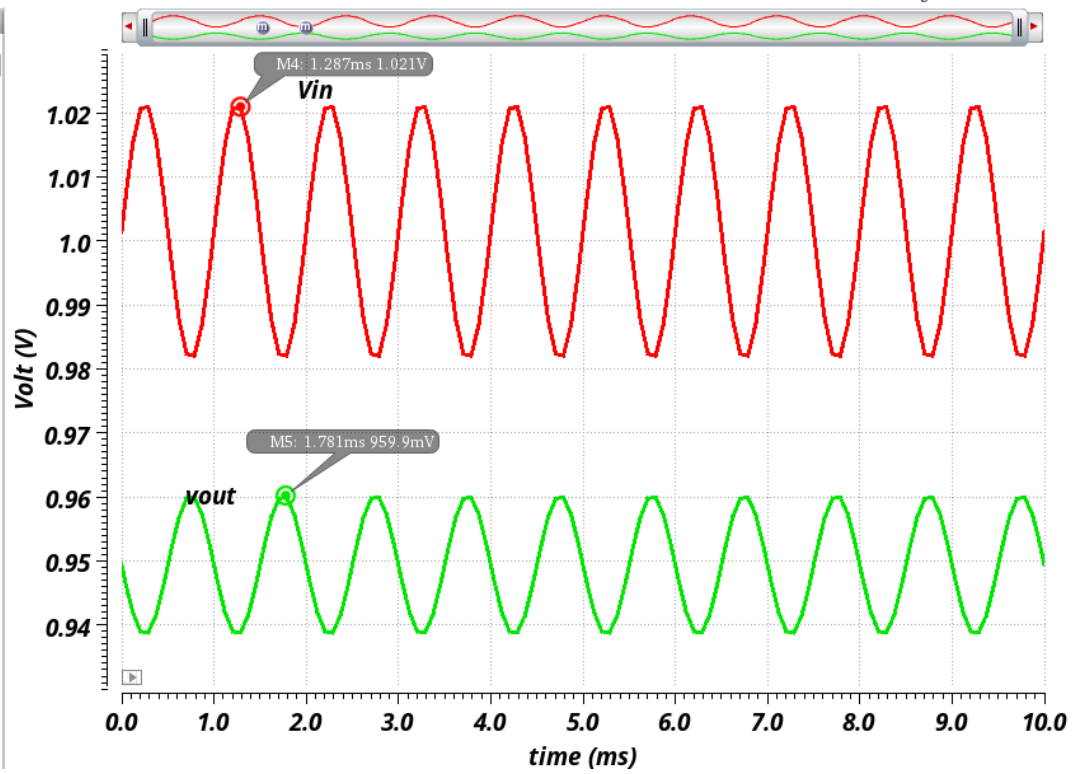
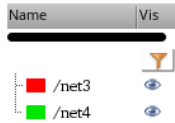
TASK 4



SCHEMATIC OF COMMON SOURCE AMPLIFIER



$V_{out}(V)$ vs $V_{in}(V)$ vs $I_d(uA)$



SMALL SIGNAL GAIN OF AMPLIFIER

a) Given power $\leq 30 \mu W$.

$$V_{DD} = 1.8.$$

$$P = V_{DD} I_{out}$$

$$I_{out} = \frac{30 \mu W}{1.8} = 16.67 \mu A.$$

$$I_{out} < 16.67$$

$$\text{Assume } I_{out} = 13 \mu A \text{ --- (1)}$$

$$V_{out} = 0.9 V.$$

$$V_{tp} = 0.5 V$$

for Pmos (diode connected load)

$$I_D = \frac{1}{2} \mu_{nlox} \left(\frac{W}{L} \right) (V_{gs} - V_{th})^2$$

$$\mu_{nlox} = 64.45 \mu A/V^2$$

$$14 \mu A = \frac{1}{2} \times (64.45) \left(\frac{W}{L} \right)_p (1.8 - 0.9 - 0.5)^2$$

$$\left(\frac{W}{L} \right) \approx 2.8$$

$$\text{Let } L = 0.18 \mu m$$

$$W = 2.8 \times 0.18 = 0.504 \mu m$$

Pmos designed.

Now for Nmos.

$$I_D = \frac{1}{2} \mu_{nlox} \left(\frac{W}{L} \right)_n (V_{gs} - V_{th})^2$$

$$\text{from task 3 } (V_{th} = 0.5) \quad \mu_{nlox} = 301 \times 10^{-6} \mu$$

$$14 \mu A = \frac{1}{2} \times 301 \times \mu \left(\frac{W}{L} \right)_n (1.8 - 0.5)^2$$

$$\left(\frac{W}{L} \right)_n \approx 0.19$$

$$L = 0.18$$

$$\omega_n = 0.19 \times 0.18 = 0.0342 \approx$$

⇒ But when we change the value $(\frac{\omega}{L})_{dd}$ to $(\frac{\omega}{L})_{new}$, we get $V_{gs} = 1.0015V$. for above values.

$$V_{gs} = 1.0015V$$

$$I_{d1} = 14 \mu A$$

$$P = I_{d1} \times V_{dd} = 1.8 \times 13.85 \mu = \underline{24.93 \mu W}$$

$$\text{gain} = \frac{V_{out}}{V_{in}} = \frac{0.9599}{1.021} \approx \underline{\underline{0.959}}$$